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OGC Observations, Measurements and Samples

—

Logical Model

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i. Keywords

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, Observations, Measurements, Samples, Monitoring Preface

The approved OGC Observations, measurements and samples (OMS) model with its three modeling layers creates undue overhead for the implementation of domain models. Thus, a logical model layer compliant with the OMS model that provides all necessary associations has been created and is described in this document. Domain models created based on the OMS logical model are compliant with OMS. In addition, concepts originating from the Environmental Monitoring Facilities Schema (EF) originating from the European INSPIRE Initiative have been integrated as a valuable complement to the OMS Host concept.

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ii. Security Considerations

No security considerations have been made for this Standard

iii. Submitting organizations

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

DataCove e.U.

Fraunhofer IOSB

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1. Scope

This document defines a logical abstract schema for observations, for features involved in the observation process, and for features involved in sampling when making observations. These provide models for the exchange of information describing observation acts and their results, both within and between different scientific and technical communities.

This model has been simplified from the multi-tier Observations, Measurements and Samples standard, flattening the modelling hierarchy to provide all OMS concepts in one consistent model. While the complexity of the full OMS model enables a great deal of flexibility in integrating concepts from existing legacy systems within OMS based models, it also proved difficult to work with as all associations referenced the conceptual interfaces in place of classes. The OMS Logical Model ameliorates these issues, easing adoption of OMS in both existing and new systems.

Observations commonly involve sampling of an ultimate feature-of-interest. This document defines a common set of sample types according to their spatial, material (for ex situ observations) or statistical nature. The schema includes relationships between sample features (sub-sampling, derived samples).

This document concerns only externally visible interfaces and places no restriction on the underlying implementations other than what is needed to satisfy the interface specifications in the actual situation.

2. Conformance

This Standard defines an abstract logical transposition of the OGC Observations, measurements and samples standard.

Conformance with this Standard shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the OGC Compliance Testing Policies and Procedures and the OGC Compliance Testing web site¹.

All requirements-classes and conformance-classes described in this document are owned by the Standard(s) identified.

¹ www.opengeospatial.org/cite

3. References

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 19103:2015, *Geographic information — Conceptual schema language*

ISO 19107:2019, *Geographic information — Spatial schema*

ISO 19108:2002, *Geographic information — Temporal schema*

ISO 19156: 2023, *Geographic information — Observation, measurements and Samples*

4. Terms and Definitions

This document used the terms defined in Policy Directive 49², which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this Standard and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

For the purposes of this document, the following additional terms and definitions apply.

4.1

application schema

conceptual schema for data required by one or more applications

[SOURCE: ISO 19101-1:2014, 4.1.2]

² https://portal.ogc.org/public_ogc/directives/directives.php

4.2

coverage

feature that acts as a function to return values from its range for any direct position within its domain

[SOURCE: ISO/DIS 19123-1, 3.1.8]

4.3

data type

specification of a value domain with operations allowed on values in this domain

EXAMPLE Integer, Real, Boolean, String and Date.

Note 1 to entry: Data types include primitive predefined types and user-definable types.

[SOURCE: ISO 19103:2015, 4.14]

4.4

domain

well-defined set

Note 1 to entry: All elements within a domain (set) are of a given type.

[SOURCE: ISO 19109:2015, 4.8, modified — Original Note 1 to entry has been replaced with a new note to entry.]

4.5

domain feature

feature of a type defined within a particular application domain

Note 1 to entry: This can be contrasted with observations and sampling features, which are features of types defined for cross-domain purposes.

4.6

ex situ

off-site

referring to the study, maintenance or conservation of a specimen or population away from its natural surroundings

Note 1 to entry: Opposite of *in situ* (on-site).

Note 2 to entry: an example of *ex situ* & direct is measuring a patient's temperature with a mercury thermometer in a blood-sample.

Note 3 to entry: an example of *ex situ* & remote is measuring a patient's temperature with an infra-red thermometer pointed at the blood sample.

4.7

feature

ion of real-world phenomena

Note 1 to entry: A feature can occur as a type or an instance. In this document, feature instance is meant unless otherwise specified.

[SOURCE: ISO 19101-1:2014, 4.1.11, modified — Note 1 to entry has been modified.]

4.8

feature-of-interest

subject of the observation

4.9

feature type

class of features having common characteristics

4.10

in situ

on-site

referring to the study, maintenance or conservation of a specimen or population without removing it from its natural surroundings

Note 1 to entry: Opposite of *ex situ* (off-site).

Note 2 to entry: an example of *in situ* & direct is measuring a patient's temperature with a mercury thermometer in the patient's rectum.

Note 3 to entry: an example of *in situ* & remote is measuring a patient's temperature with an infra-red thermometer at a distance.

4.11

measure

<GML> value described using a numeric amount with a scale or using a scalar reference system

Note 1 to entry: When used as a noun, measure is a synonym for physical quantity.

[SOURCE: ISO 19136-1:2020, 3.1.41]

4.12

measurement

set of operations having the object of determining the value of a quantity

[SOURCE: ISO 19101-2:2018, 3.21]

4.13

observation

act carried out by an observer to determine the value of an observable property of an object (feature-of-interest) by using a procedure, with the value is provided as the result

4.14

observation result

estimate of the value of a property determined through a known observation procedure

[SOURCE: ISO 19156:2011, 4.14]

4.15

observer

identifiable entity that can generate observations pertaining to an observable property by implementing a procedure

Note 1 to entry: An observer is an instance of a sensor, instrument, implementation of an algorithm or a being such as a person.

4.16

procedure

specified way to carry out an activity or a process

[SOURCE: ISO 9000:2015, 3.4.5, modified — Note 1 to entry has been deleted.]

4.17

process

set of interrelated or interacting activities that use inputs to deliver an intended result

[SOURCE: ISO 9000:2015, 3.4.1, modified — Notes 1-6 have been deleted.]

4.18

property

facet or attribute of an object referenced by a name

EXAMPLE Abby's car has the colour red, where "colour red" is a property of the car.

Note 1 to entry: in some communities, the observed property is referred to as the measurand

[SOURCE: ISO 19143:2010, 4.21, modified — Example and note have been added to the entry.]

4.19

property type

characteristic of a feature type

EXAMPLE Cars (a feature type) all have a characteristic colour, where "colour" is a property type.

Note 1 to entry: The value for an instance of an observable property type can be estimated through an act of observation.

Note 2 to entry: In chemistry-related applications, the term "determinand" or "analyte" is often used.

Note 3 to entry: Adapted from ISO 19109:2005.

4.20

proximate feature-of-interest

entity that is directly of interest in the act of observing

Note 1 to entry: This is a specialized form of the feature-of-interest

4.21

range

<coverage> set of feature attribute values associated by a function, the coverage, with the elements of the domain of a coverage

Note 1 to entry: This is consistent with the more generic definition of range in [ISO 19107:2019](#).

[SOURCE: [ISO/DIS 19123-1](#), 3.1.47]

4.22

sample

object that is representative of a concept, real-world object or phenomenon

4.23

sampler

device or entity (including humans) that is used by, or implements, a sampling procedure to create or transform one or more sample(s)

4.24

sensor

element of a measuring system that is directly affected by a phenomenon, body, or substance carrying a quantity to be measured

[SOURCE: JCGM 200:2012, 3.8, modified — EXAMPLES and NOTE deleted.]

4.25

ultimate feature-of-interest

entity that is ultimately of interest in the act of observing

Note 1 to entry: This is a specialized form of the feature-of-interest.

4.26

unit of measure

reference quantity chosen from a unit equivalence group

Note 1 to entry: In positioning services, the usual units of measurement are either angular units or linear units. Implementations of positioning services must clearly distinguish between SI units and non-SI units. When non-SI units are employed, it is required that their relation to SI units be specified.

[SOURCE: ISO 19116:2019, 3.29]

4.27

value

element of a type domain

Note 1 to entry: A value considers a possible state of an object within a class or type (domain).

Note 2 to entry: A data value is an instance of a datatype, a value without identity.

Note 3 to entry: A value can use one of a variety of scales including nominal, ordinal, ratio and interval, spatial and temporal. Primitive datatypes can be combined to form aggregate datatypes with aggregate values, including vectors, tensors and images.

[SOURCE: ISO/IEC 19501:2005, 0000_5, modified — Note 3 to entry has been added.]

5. Conventions

5.1 Abbreviated terms and acronyms

EO	Earth observation
GFM	General Feature Model
GML	Geography Markup Language
INSPIRE	Infrastructure for Spatial Information in Europe

O&M	Observations and Measurements
OMS	Observations, Measurements and Samples (this current version of ISO 19156)
OGC	Open Geospatial Consortium
SensorML	OGC Sensor Model Language
SOS	OGC Sensor Observation Service
STA	OGC SensorThings API
SWE	OGC Sensor Web Enablement
UML	Unified Modeling Language
XML	Extensible Markup Language
2-D	two dimensional
3-D	three dimensional

5.2 Schema language

The conceptual schema specified in this document is in accordance with the Unified Modelling Language (UML) ISO/IEC 19501, following the guidance of ISO 19103.

The UML in Abstract Core and Basic packages is conformant with the profile described in ISO 19136-1:2020, Annex E. Use of this restricted idiom supports direct transformation into a GML Application Schema. The stereotype «FeatureType» states that a class is an instance of the «metaclass» FeatureType (ISO 19109) [2], and therefore represents a feature type.

The prose explanation of the model uses the term “property” to refer to both class attributes and association roles. This is consistent with the General Feature Model described in ISO 19109. In the context of properties, the term “value” refers to either a literal (for attributes whose type is simple), or to an instance of the class providing the type of the attribute or target of the association. Within the explanation, the property names (property types) are sometimes used as natural language words where this assists in constructing a readable text.

5.3 Model element names

This document specifies a model for observations using terminology that is based on current practice in a variety of scientific and technical disciplines. It is designed to apply across disciplines, so the best or “most neutral” term has been used in naming the classes, attributes and associations provided. The terminology does not, however, correspond precisely with any single discipline.

5.4 Identifiers

The normative provisions in this specification are denoted by the URI

<http://www.opengis.net/spec/{standard}/{m.n}>

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

6. Applying this model (informative)

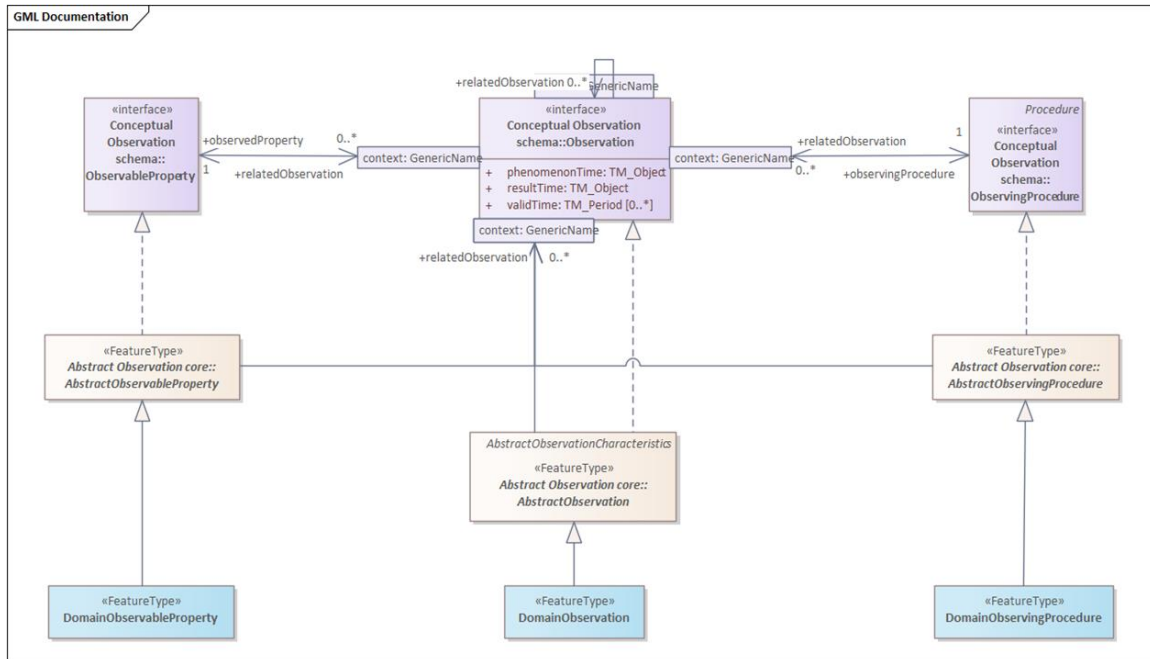
6.1 General

The approved OMS model with its three modeling layers creates undue overhead for the implementation of domain models. Thus, a logical model layer compliant with the OMS model that provides all necessary associations has been created and is described in this document.

OMS provides the relevant concepts for the structured description of observations, including the sampling structure, often essential for true understanding of the nature of the observations being provided. As data provisioning mechanisms are transitioning towards highly distributed linked approaches, the model structure and packaging have been significantly abstracted. This approach allows implementers to explicitly select the concepts to be supported based on their requirements, while clearly stating to which requirements and Conformance Classes their implementation complies. Both the Observation and Sample sections of this model have been structured using the following layering of packages:

- **Conceptual:** Within the Conceptual Model Packages, only Interfaces are provided. These models provide a very abstract view of the individual concepts they contain without reference to specific implementations. This approach allows for the inclusion of semantically aligned objects from external sources, that while not having been created under the Observations, measurements and samples model, do provide concepts sharing the same semantic meaning as the concepts from the Conceptual Models.
- **Abstract Core:** Within the Abstract Core Model Packages, only abstract featureTypes are provided following the semantic structure of the Conceptual model (i.e., realizing the interfaces provided by the Conceptual Model Packages). A consistent approach to metadata provision is introduced. All associations from the abstract featureTypes reference the conceptual interfaces for greater implementation flexibility. The Abstract Core Model Packages are foreseen for the creation of domain models, as required by individual communities, providing an Abstract Core ready for extension. Due to their abstract definition, the classes from Abstract Core cannot be directly instantiated.
- **Basic:** Within the Basic Packages, simple concrete featureTypes (specializing the abstract ones from the Abstract Core model) have been defined with some basic utility attributes added for rapid out-of-the-box deployment. A few additional concepts pertaining to collections and potential observations are introduced at this level. This is the only level of OMS that can be directly instantiated.

While the layered approach described above allows for great flexibility, as required of a conceptual model, it also adds a serious burden on those tasked with specializing parts of OMS within their domain models. As part of the flexibility of the OMS layered approach, all associations across model layers target the interfaces from the conceptual model. In order to derive domain classes from the abstract classes provided by OMS, one must also include the interfaces for relevant context on associations. This is partially due to Enterprise Architect (the UML modeling tool used) not allowing to display inherited associations as can be done with inherited attributes. The resulting diagrams rapidly become unreadable, as shown in the simple example below.



In this example, we are specializing the Observation, ObservableProperty and ObservingProcedure types to specialized domain types (e.g. because additional attributes are required). The top third of the diagram only serves the purpose of showing the associations between the interfaces This adds additional unwanted overhead to models.

This issue was initially encountered during work done on updating the World Meteorological Organization (WMO) WIGOS Metadata Representation (WMDR) model to OMS, where it rapidly became apparent that the already complex WMDR models could not cope with the addition of the OMS interfaces together with the abstract classes. During the Time Series Modelling Language (TSML) update we encountered the same issue, and are thus basing our TSML work on the OMS Logical Model.

To overcome this, in the OMS Logical Model, all classes from all 3 layers of the OMS model are therefore provided as featureTypes, and all associations are available directly between these classes. Like the OMS model, it provides individual packages for Observation and Sampling. All these classes are aligned with the OMS model. However, not all classes are available at the conceptual level, e.g. the individual sample types SpatialSample, MaterialSample and StatisticalSample are not introduced until the OMS

Basic Samples package. In these cases, the abstract class in the logical model had to be derived from the corresponding Basic Package.

6.2 Working with FeatureTypes

Within the OMS Logical model, we make use of the GML FeatureType Stereotype for all core classes. Thus, all classes of type FeatureType natively include the following basic attributes, in addition to geometry information:

- `gml:id`: A mandatory unique identifier for each feature instance, essential for referencing and linking between features. It must be unique within the scope of the XML document.
- `gml:name`: A label or name assigned to the feature, which may or may not be unique.
- `gml:description`: A text description of the feature.

Thus, these attributes are not modelled within this standard

6.3 Creating Domain Classes

The OMS Logical model has been created for the further derivation of bespoke domain models, tailoring the OMS structure towards specific requirements both by extending the existing classes as well as by defining additional classes and associations.

While the most straightforward approach is direct alignment, including one-to-one mapping of classes while maintaining class and attribute names, depending on the use case and target technology, modifications may be required. In some cases, classes will be renamed to better reflect domain concepts, while in other cases, classes may be merged if there is a clear one-to-one association between them.

This can be illustrated by the alignment of the OGC SensorThings API (STA) data model, shown in Figure 1, with OMS, shown in Figure 2 and Figure 3. At first glance, the models seem similar but different. However, with the following alignment, it becomes clear that STA fully conforms to OMS:

- Attribute renaming: `OMS:Observation.parameter` is renamed `STA:Observation.properties`
- Class renaming: `OMS:Host` is renamed `STA:Thing`; `OMS:Observer` is renamed `STA:Sensor`
- Class merge: `OMS:ObservationCollection` and `OMS:ObservationCharacteristics` have been merged to `STA:Datastream`

In this way, the OMS model can serve as the basis for a wide range of domain models, each tailored to the specific needs of their community, while maintaining structural interoperability.

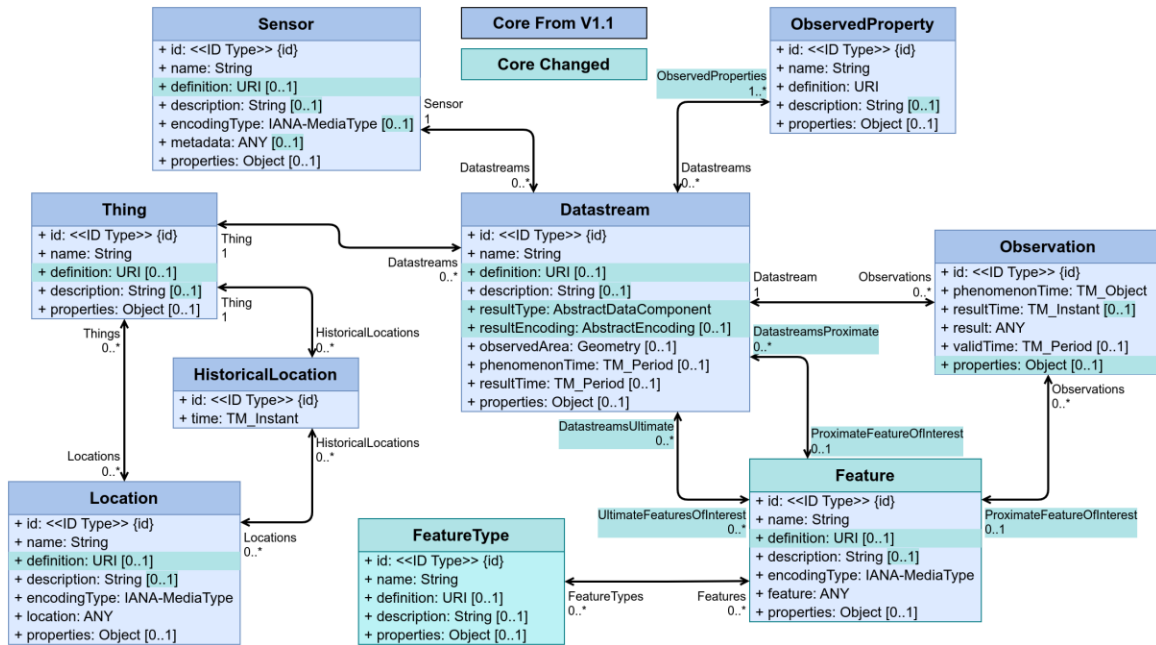


Figure 1: OGC SensorThings API data model

7. Fundamental characteristics of observations and samples (informative)

7.1 Observation Schema

7.1.1 General

The OMS Observation provides an efficient standardized method for providing information on properties of features. While the simplest approach for providing such feature information remains the definition of a simple attribute on this feature carrying this property, this approach is severely limited towards the provision of additional relevant meta-information such as:

- The measurement methodology used in determining this property;
- The measurement device that provided the value;
- The facility at which this value was determined;
- The deployment time of the measurement device at the facility;
- Semantic details on the property being provided;
- Temporal support for properties that vary over time;
- Information on related observations.

In the process of defining specialized feature models for specific domains, it is valuable to reflect if this additional observational meta-information is of relevance for individual feature properties. When feature properties are static, observational meta-information not

relevant, a simple attribute will suffice; in situations where observational meta-information is relevant, e.g., environmental monitoring and reporting, utilization of the OMS Observation for provision of this feature property greatly eases professional data sharing.

7.1.2 Observation

An **Observation** is defined as an act carried out by an **Observer** to determine the value of an **ObservableProperty** of an object (**featureOfInterest**) by using an **ObservingProcedure**; the value is provided as the **result**. It is important to note that the terms ‘observation’, ‘interpretation’, ‘forecast’, ‘simulation’ correspond to this definition. There are three different temporal properties on Observations

- **phenomenonTime**: The time for which the **result** applies to the characteristic of the **FeatureOfInterest** being observed. The **phenomenonTime** is often the time of interaction with a real-world feature either by a **SamplingProcedure** (time at which a **Sample** has been taken) or by an **ObservingProcedure**. If the result is the average of multiple samples taken at different times, then the **phenomenonTime** is the time interval over which these measurements were taken.
- **resultTime**: The instant of time when the **result** of the **Observation** became available. The **resultTime** typically corresponds to when the **Procedure** associated with the **Observation** was completed. For some observations this is identical to the **phenomenonTime**. However, there are important cases where they differ:
 - Where a measurement is made on a specimen in a laboratory, the **phenomenonTime** is the time the specimen was retrieved from its host, while the **resultTime** is the time the laboratory procedure was applied.
 - Where sensor observation results are post-processed, the **resultTime** is the post-processing time, while the **phenomenonTime** is the time of initial interaction with the world.
 - Simulations may be used to estimate the values for phenomena in the future or past. The **phenomenonTime** is the time that the result applies to, while the **resultTime** is the time that the simulation was executed.
- **validTime**: The time interval during which the **result** is assumed to be applicable for use. This attribute is commonly required in forecasting applications.

The **featureOfInterest** (FoI) association links to the feature that is the subject of the **Observation**. While in many cases, the **FoI** is a **Sample** (see section 7.2.2), any feature can serve as **FoI**. This object is either the real-world object whose properties are under observation, or it is an object used as a proxy for a real-world object that is not directly observable. Some examples of **FoIs** are:

- An instance of a feature modelled in a specific domain model (e.g., Borehole according to OGC GeoSciML).
- The bubble of air around the intake of an air quality monitoring station.
- An existing well being used for water quality measurements.

In some cases, the measurement process may be performed on an intermediary entity referred to as **proximateFeatureOfInterest** that acts as a proxy to the ultimate feature-of-interest that is being observed (measured, estimated or calculated). Thus, within the OMS model, there is a differentiation between:

- **proximateFeatureOfInterest**: The entity that is directly of interest in the act of observing.
- **ultimateFeatureOfInterest**: The entity that is ultimately of interest in the act of observing.

Some examples to illustrate the differentiation between proximate and ultimate FoI:

- To determine the concentrations of chemical compounds in a river, a sample is taken in a predefined location in the river. This sample is taken to a laboratory where the required chemical analysis is done. In this case, the river is the **ultimateFeatureOfInterest**, while the sample is the **proximateFeatureOfInterest**.
- Pertaining to documents and observations on the consistency thereof, for the Observation “This clause is inconsistent”, the **ultimateFeatureOfInterest** is the entire document, while the **proximateFeatureOfInterest** is the specific clause being addressed.

Observational meta-information is provided by the following associations

- The **observedProperty** association links to the **ObservableProperty** that is the subject of the **Observation**. The **ObservableProperty** referenced by **observedProperty** should correspond to a characteristic associated with the **featureOfInterest**.
- The **observingProcedure** association links to the **ObservingProcedure** used by the **Observation** to determine the value of the **ObservableProperty** provided by the **result**. The **ObservingProcedure** referenced by **procedure** should be suitable for the associated **ObservableProperty**.
- The **observer** association links to the **Observer** that is involved in the creation of this **Observation**.
- The **host** association links to the **Host** that is involved in the creation of this **Observation**. At least one **Observer** or **Host** should be provided.

Finally, the **result** association links to the **result** of the **Observation**. The type of the result provided by the **result** association should be suitable for the associated **ObservableProperty**

The **Observation** will provide a unit of measure (UoM) if the result is measurable. If the UoM is not contained in the result, it shall be provided in the context of the **Observation**; the provision modality is to be defined by communities. The unit of measure should be suitable for the associated **ObservableProperty** and **ObservingProcedure**. In the case where the result of the Observation is a classification, for which no unit exists, the UoM can be declared as unitless

The **parameter** attribute provides a flexible extension point, allowing for the provision of name/value pairs further describing the **Observation**. **Parameter** should not be utilized to provide information already contained in the model by existing attributes or associations.

7.1.3 ObservableProperty

An **ObservableProperty** is defined as a quality (property, characteristic) of the **feature-of-interest** that can be observed. The height of a tree, the depth of a water body, or the temperature of a surface are examples of observable properties, while the value of a classic car is not (directly) observable but asserted.

The association **observer** links to an **Observer** capable of observing this **ObservableProperty**.

7.1.4 ObservingProcedure

An **ObservingProcedure** is defined as the description of steps performed in order to determine the value of an **observableProperty** by an **Observer**.

Depending on the complexity of the use case, the **ObservingProcedure** will be more or less explicitly described. Especially pertaining to historical data, there may be very little or no information available – what information is available should be provided;

The **ObservingProcedure** is a workflow, protocol, plan, algorithm, or computational method specifying how to make an observation. The **ObservingProcedure** is often referred to as the method, or the recipe that the observer (cook) follows to generate the **Observation**. Different **Observers** can follow the same (reusable) **ObservingProcedure** for the creation of different **Observations**.

The **ObservingProcedure** cannot describe a sensor instance, but it can describe the sensor type.

An instance of **ObservingProcedure** is a description of the process utilized by an **Observer**, this could be a chemical analysis method, a protocol for measuring an object, but could also be a checklist utilized by a human **Observer** during a biodiversity campaign. **ObservingProcedure** could further describe the algorithms behind simulators or models used to generate a result from other inputs.

The association **observer** links to an **Observer** capable of performing this **ObservingProcedure**.

7.1.5 Observer

An **Observer** is defined as an identifiable entity that can generate **Observations** pertaining to an **observableProperty** by implementing an **ObservingProcedure**.

The **Observer** is the entity instance of a sensor, instrument, implementation of an algorithm, or a being such as a person, not the entity type. It responds to a stimulus, e.g., a change in the environment, or input data composed from the results of prior **Observations**, and generates a **result**.

Pertaining to sensors, the **Observer** would reference the explicit sensor, while the **ObservingProcedure** would reference the methodology utilized by that sensor type. An **Observer** can be hosted by one or more **Host**. Different **Observers** can follow the same (reusable) observing **ObservingProcedure** for the creation of different **Observations**. An **Observer** is closely linked with an **ObservableProperty** that it generates **results** for;

Accelerometers, gyroscopes, barometers, magnetometers, and so forth are **Observers** that are typically mounted on a modern smartphone (which acts as **Host**). Other examples of **Observers** include the human eyes.

Contextual information to the **Observer** is provided via the following associations:

- The **observableProperty** association links to an **ObservableProperty** that this **Observer** can observe.
- The **observingProcedure** association links to an **ObservingProcedure** that this **Observer** can perform.
- The **deployment** association links to a **Deployment** to which this **Observer** is either physically or organizationally attached.

7.1.6 Host

A **Host** is defined as a grouping of **Observers** for a specific reason.

In many use cases, the **Host** is the environmental monitoring facility. The **Host** can be a platform that hosts a set of sensors. An alternative usage could pertain to a biodiversity survey campaign. In this scenario, the team performing the survey would be modelled as observers whereas the entire survey campaign can be represented as a **Host**.

The association **deployment** links to a **Deployment** at this **Host**.

The association **relatedHost** links to a **Host** the **Host** is related to.

7.1.7 Deployment

A **Deployment** is defined as information on the assignment of an **Observer** to a **Host**. Examples of **Deployment** are:

- Information regarding a sensor being attached to a pole.
- The monitoring facilities pertaining to an environmental monitoring network.
- The description of a ship cruise linking a research vessel with a marine network.

- The participation of a citizen in a citizen-science project involving crowd sensing.

The attribute **deploymentReason** provides a human readable description of the reason for the **Deployment**. The attribute **deploymentTime** provides the time that the **Deployment** pertains to.

Contextual information to the **Deployment** is provided via the following associations:

- The **observer** association links to the **Observer** associated with this **Deployment**.
- The **host** association links to the **Host** to which this **Deployment** pertains.

7.1.8 ObservationCharacteristics

An **ObservationCharacteristics** is defined as a set of common characteristics used for describing an **Observation** or a collection of **Observations**. **ObservationCharacteristics** has all of the attributes and associations as **Observation**, but none of the cardinality limitations (all attributes and associations have cardinality 0..*), making it a useful concept when describing observational matters.

7.1.9 ObservationCollection

An **ObservationCollection** is defined as a collection of similar **Observations**. The **ObservationCollection** is further described by the associated **ObservationCharacteristics** (linked via the **memberCharacteristics** association), that can carry all observational meta-information directly provided by **Observations**.

The **collectionType** indicates how the linked **memberCharacteristics** impact the **Observations** contained within the **ObservationCollection**. The following values are foreseen for **collectionType**:

- **homogeneous**: all observations contained are of a similar nature.
If a property value is provided within the **ObservationCharacteristics**, this value applies to all **Observations** contained in the **ObservationCollection**. The **Observations** need not contain attributes or associations supplied via the **ObservationCharacteristics**:
 - property not provided – values may be provided by the **Observations** but is not provided at this level
 - property provided but with no content – no **Observation** within the collection provides this property
 - property = value – this value applies to all **Observations** within the collection
 - property = value set/range – this value set/range applies to all **Observations** within the collection

- **summarizing**: a wider grab-bag type of collection.
If multiple values for a property are available in the contained **Observations**, all values for this attribute (or the range of values contained in all **Observations**) are provided in the **ObservationCharacteristics**. A property may also be empty in the **ObservationCharacteristics** – in this case any value can be provided for this attribute within the contained **Observations**:
 - property not provided – values may be provided by the **Observations** but a summary is not provided at this level;
 - property provided but with no content – no **Observation** within the collection provides this property;
 - property = value – this value applies to all **Observations** within the collection;
 - property = value set/range – all **Observations** provide a value within this set/range.

The **member** association links to all **Observation(s)** contained within the **ObservationCollection**.

The **relatedCollection** association links to an **ObservationCollection** related with this **ObservationCollection**.

7.2 Sample Schema

7.2.1 General

The OMS **Sample** allows for provision of information of **Samples**, specifically for use as the **FoI** of an **Observation**. It enables provision of relevant meta-information on how **Samples** were derived and processed, including:

- The sampling act that resulted in the sample;
- The device used in gaining the sample;
- The procedure followed in gaining the sample;
- Individual preparation steps performed on the sample prior to measurement;
- The procedures followed in the individual preparation steps.

In addition, different types of **Samples** have been specified:

- Spatial samples have a geospatial component;
- Material samples are physical tangible samples, often referred to as specimens.
- Statistical samples

7.2.2 Sample

A **Sample** is defined as an object that is representative of a concept, real-world object or phenomenon. Typically, the **Sample** is a **Feature** which is intended to be representative of a **FeatureOfInterest** on which **Observations** can be made. As such, it can carry a characteristic pertaining to the **ObservedProperty** being evaluated by the **Observation**.

The way the **Sample** is taken is typically guided by a sampling strategy. **Samples** are often artefacts of an observational strategy, and often have no significant function outside of their role in the observation process (although specimen preservation could be considered a specific activity per se).

For example, a profile typically samples a water- or atmospheric-column; a well samples the water in an aquifer; a tissue specimen samples a part of an organism. A statistical sample is often designed to be characteristic of an entire population, so that **Observations** can be made regarding the sample that provide a good estimate of the properties of the population.

The **sampleType** attribute can be used to indicate the type of **Sample** according to a community agreed typology.

Contextual information to the **Sample** is provided via the following associations:

- The **sampling** association links to the **Sampling** act the **Sample** is the result of;
- The **sampledFeature** association links to the feature the **Sample** is intended to be representative of;
- The **preparationStep** association links to the **PreparationStep(s)** applied to prepare the **Sample**;
- The **relatedSample** association links to a **Sample** the **Sample** is related to.

The **parameter** attribute provides a flexible extension point, allowing for the provision of name/value pairs further describing the **Sample**. **Parameter** should not be utilized to provide information already contained in the model by existing attributes or associations.

7.2.3 SpatialSample

A **SpatialSample** is defined as a geospatial **Sample**. When **Observation(s)** are made to estimate properties of a geospatial feature, in particular where the value of a property varies within the scope of the feature, a **SpatialSample** is used. Depending on accessibility and on the nature of the expected property variation, the **SpatialSample** can be extensive in one, two or three spatial dimensions.

Some common names for **SpatialSample** used in various application domains include Borehole, Flightline, Interval, Lidar Cloud, Map Horizon, Microscope Slide, Mine Level,

Mine, Observation Well, Profile, Pulp, Quadrat, Scene, Section, ShipsTrack, Spot, Station, Swath, Trajectory, Traverse, etc.

The following attributes provide relevant spatial information:

- The **shape** attribute provides the Geometry of the **SpatialSample**;
- The **horizontalPositionalAccuracy** attribute provides information on the positional accuracy of the horizontal component of the Geometry of the **SpatialSample**;
- The **verticalPositionalAccuracy** attribute provides information on the positional accuracy of the vertical component of the Geometry of the **SpatialSample**.

7.2.4 MaterialSample

A **MaterialSample** is defined as a physical, tangible **Sample**. **MaterialSample(s)** that are curated and preserved are sometimes known as ‘specimens’. **MaterialSample(s)** can be destroyed in connection with the observation act. A **MaterialSample** is a physical **Sample** of a **FeatureOfInterest**, obtained for **Observation(s)** normally carried out *ex situ*, sometimes in a laboratory.

The following attributes provide relevant information:

- The **size** attribute provides describes a physical extent of the **MaterialSample**. The size can be length, mass, volume, etc., as appropriate for the **MaterialSample** instance and its material type;
- The **storageLocation** attribute provides the location of a **MaterialSample**. The **storageLocation** can be a location such as a shelf in a warehouse or a drawer in a museum;
- The **sourceLocation** attribute provides the location from where the **MaterialSample** was obtained. The attribute **sourceLocation** of the **MaterialSample** can be unnecessary in the case the related **Sampling** act **samplingLocation** attribute is provided or when a relatedSample’s location provides an unambiguous location.

7.2.5 StatisticalSample

A **StatisticalSample** is defined as a statistical subset of a feature-of-interest, defined for the purpose of creating **Observation(s)**. **StatisticalSample(s)** usually apply to populations or other sets, of which certain subset can be of specific interest, e.g., the male or female subset of a population.

The following attributes provide relevant information:

- The **classification** attribute describes a criterion by which the subset was defined. The classification can be age, gender, etc., as appropriate for the set or population

on which the subsetting is performed. A **StatisticalClassification** is provided for this purpose with the attributes:

- The **concept** attribute defines the concept by which a **StatisticalClassification** is to be performed, e.g., age, gender, color, size etc.;
- The **classification** attribute provides the explicit classification classes pertaining to the classification concept described by the **StatisticalClassification**, e.g., Age Brackets: [0-10], [10-20]; Genders: Male, Female, Other; Color: Red, Green, Blue.

7.2.6 Sampling

A **Sampling** is defined as an act applying a **SamplingProcedure** to create or transform one or more **Sample(s)**. Examples of **Sampling** are:

- Crushing a rock sample in a ball mill;
- Digging a pit through a soil sequence;
- Dividing a field site into quadrants;
- Drawing blood from a patient;
- Extracting water from an observation well;
- Extracting a sample from a defined environmental monitoring station;
- Registering an image of the landscape;
- Sieving a powder to separate the subset finer than 100-mesh;
- Selecting a subset of a population;
- Splitting a piece of drill-core to create two new samples;
- Taking a diamond-drill core from a rock outcrop.

The **samplingLocation** attribute can be used to provide location information pertaining to the **Sampling** act. The time attribute provides information on the time of the **Sampling**.

Contextual information to the **Sampling** act is provided via the following associations:

- The **sample** association links to the **Sample** generated by the **Sampling**;
- The **featureOfInterest** association links to the concept, real-world object or phenomenon (feature-of-interest) the **Sample(s)** of the **Sampling** represent;
- The **sampler** association links to the **Sampler** that performed the **Sampling**;
- The **samplingProcedure** association links to the **SamplingProcedure** used by the **Sampling**;
- The **relatedSampling** association links to related **Sampling(s)**.

The **parameter** attribute provides a flexible extension point, allowing for the provision of name/value pairs further describing the **Sampling** act. For example, when taking water samples, the sampling procedure specifies that an amount of time needs to pass to allow sediments to settle. The exact waiting time used in this Sampling can be stored in the parameter.

Parameter should not be utilized to provide information already contained in the model by existing attributes or associations.

7.2.7 Sampler

A **Sampler** is defined as a device or entity (including humans) that is used by, or implements, a **SamplingProcedure** to create or transform one or more **Sample(s)**. Examples of **Samplers** include:

- A ball mill, diamond drill, hammer;
- A hypodermic syringe and needle;
- An image sensor, a soil auger;
- A human being.

All the examples above can act as sampling devices (i.e., be **Samplers**). However, sometimes the distinction between the **Sampler** and the sensor (**Observer**) is not evident, as they are packaged as a unit. A **Sampler** need not be a physical device.

The **samplerType** attribute can be used to indicate the type of **Sampler** according to a community agreed typology.

Contextual information to the **Sampler** is provided via the following associations:

- The **sampling** association links to the **Sampling** act performed by the **Sampler**;
- The **implementedProcedure** association links to the **SamplingProcedure** implemented by the **Sampler**.

7.2.8 SamplingProcedure

A **SamplingProcedure** is defined as the description of steps performed by a **Sampler** in order to extract a **Sample** from its **sampledFeature** in the frame of a **Sampling**.

Contextual information to the **SamplingProcedure** is provided via the following associations:

- The **sampling** association can be used to link to the **Sampling** act(s) where this procedure has been used;
- The **sampler** association can be used to link to **Sampler(s)** that implement this procedure.

7.2.9 PreparationStep

A **PreparationStep** is defined as an individual step pertaining to a **PreparationProcedure**.

The **description** attribute can be used to provide a human readable description of the **PreparationStep**. The **time** attribute indicates the time the **PreparationStep** was performed.

Contextual information to the **PreparationStep** is provided via the following associations:

- The **processingDetails** association links to a **PreparationProcedure** step performed on the **Sample** the **PreparationStep** pertains to;
- The **preparedSample** association links to the **Sample** on which the **PreparationProcedure** is performed.

7.2.10 PreparationProcedure

A **PreparationProcedure** is defined as the description of preparation steps performed on a **Sample**.

The **samplePreparationStep** association can be used to link to the **PreparationStep** where this **PreparationProcedure** has been used.

7.2.11 SampleCollection

A **SampleCollection** is defined as a collection of **Sample(s)**.

The **member** association links to all **Sample(s)** contained within the **SampleCollection**.

The **relatedCollection** association links to an **SampleCollection** related with this **SampleCollection**.

7.3 Environmental Monitoring Facilities Schema

7.3.1 General

The Environmental Monitoring Facilities Schema (EF) originates from the European INSPIRE Initiative³ underpinned by the INSPIRE Directive (2007/2/EC⁴). When EF was originally defined, it was based on O&M, the predecessor to OMS; in this version, it has been fully aligned with the OMS Logical Model. As the concepts covered in EF are applicable to a wide range of environmental monitoring undertakings, a core audience of the OMS standard, they are being brought to a wider audience in the framework of this standard.

The EF schema covers the following concepts:

- **Environmental Monitoring Facilities**, installations/locations where environmental measurements take place, usually over time;
- **Environmental Monitoring Networks**, describing administrative or organizational groupings of Environmental Monitoring Facilities;
- **Environmental Monitoring Programmes** that define frameworks based on policy relevant documents defining the target of a collection of observations;
- **Environmental Monitoring Activities**, circumscribing targeted activities in a coherent and concise timeframe, area and purpose.

Within the EF Schema sections of this document, liberal reuse of original INSPIRE texts has been made.

7.3.2 MonitoringFacility

A georeferenced object directly collecting or processing data about objects whose properties (e.g. physical, chemical, biological or other aspects of environmental conditions) are repeatedly observed or measured. An environmental monitoring facility can also host other environmental monitoring facilities.

Laboratories are not EnvironmentalMonitoringFacilities as the exact location of the laboratory does not add further information to the measurement. The methodology used in the laboratory should be provided with observational data.

The following attributes further characterize the **MonitoringFacility**:

- **id**: Identifier;
- **name**: Plain text denotation of the **MonitoringFacility**;

³ https://knowledge-base.inspire.ec.europa.eu/index_en

⁴ <https://eur-lex.europa.eu/eli/dir/2007/2/oj/eng>

- **additionalDescription**: Plain text description of additional information not fitting in other attributes;
- **mediaMonitored**: Monitored environmental medium;
- **legalBackground**: The legal context, in which the management and regulation of the **MonitoringFacility** is defined;
- **responsibleParty**: Responsible party for the **MonitoringFacility**;
- **geometry**: Geometry associated to the **MonitoringFacility**. For mobile facilities the geometry represents the area the facility is expected to measure in;
- **onlineResource**: A link to an external document providing further information on the **MonitoringFacility**;
- **purpose**: Reason for which the **MonitoringFacility** has been generated;
- **reportedTo**: Information on the involvement of the **AbstractMonitoringFeature** in reporting;
- **representativePoint**: Representative location for the **MonitoringFacility**;
- **measurementRegime**: Regime of the measurement;
- **mobile**: Indicate whether the **MonitoringFacility** is mobile (repositionable) during the acquisition of the observation;
- **resultAcquisitionSource**: Source of result acquisition;
- **specialisedEMFType**: Categorization of **MonitoringFacility(s)** generally used by domain and in national settings.
Examples: platform, site, station, sensor.

In addition, the following associations provide contextual information:

- **observingCapability**: A link pointing to the explicit capability of an **MonitoringFacility**. This provides a clean link between the observed property, the procedure used as well as the location of the measurement;
- **broader**: A link pointing to a broader **MonitoringFacility** (a higher level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;
- **narrower**: A link pointing to narrower **MonitoringFacility(s)** (a lower level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;
- **supersedes**: In a genealogy, the **MonitoringFacility(s)** that has(have) been deactivated/replaced by another one;
- **supersededBy**: In a genealogy, the newly active **MonitoringFacility(s)** that replaces(replace) the superseded one;
- **involvedIn**: **MonitoringActivity(s)** in which the **MonitoringFacility** is involved;
- **belongsTo**: A link pointing to the **MonitoringNetwork(s)** this **MonitoringFacility** pertains to. The association has additional properties as defined in the association class **NetworkFacility**;
- **operationalActivityPeriod**: Lifespan of the physical object (facility);

- **relatedTo:** Any Thematic Link to an Environmental Monitoring Facility. The association has additional properties as defined in the association class **AnyDomainLink**.

7.3.3 MonitoringNetwork

Administrative or organisational grouping of **MonitoringFacility(s)** managed the same way for a specific purpose, targeting a specific area. Each network respects common rules aiming at ensuring coherence of the observations, especially for purposes of **MonitoringFacility(s)**, mandatory parameters selection, measurement methods and measurement regime.

The following attributes further characterize the **MonitoringNetwork**:

- **id:** Identifier;
- **name:** Plain text denotation of the **MonitoringNetwork**;
- **additionalDescription:** Plain text description of additional information not fitting in other attributes;
- **mediaMonitored:** Monitored environmental medium;
- **legalBackground:** The legal context, in which the management and regulation of the **MonitoringNetwork** is defined;
- **responsibleParty:** Responsible party for the **MonitoringNetwork**;
- **geometry:** Geometry associated to the **MonitoringNetwork**. For mobile facilities the geometry represents the area the facility is expected to measure in;
- **onlineResource:** A link to an external document providing further information on the **MonitoringNetwork**;
- **purpose:** Reason for which the **MonitoringNetwork** has been generated;
- **reportedTo:** Information on the involvement of the **AbstractMonitoringFeature** in reporting;
- **organisationLevel:** Level of legal organisation the **MonitoringNetwork** is affiliated with.

In addition, the following associations provide contextual information:

- **observingCapability:** A link pointing to the explicit capability of an **MonitoringNetwork**. This provides a clean link between the observed property, the procedure used as well as the location of the measurement;
- **broader:** A link pointing to a broader **MonitoringNetwork** (a higher level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;
- **narrower:** A link pointing to narrower **MonitoringNetwork(s)** (a lower level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;

- **supersedes**: In a genealogy, the **MonitoringNetwork(s)** that has(have) been deactivated/replaced by another one;
- **supersededBy**: In a genealogy, the newly active **MonitoringNetwork(s)** that replaces(replace) the superseded one;
- **involvedIn**: **MonitoringActivity(s)** in which the **MonitoringNetwork** is involved;
- **contains**: A link pointing to the **MonitoringFacility(s)** included in this **MonitoringNetwork**. The association has additional properties as defined in the association class **NetworkFacility**.

7.3.4 MonitoringProgramme

Framework based on policy relevant documents defining the target of a collection of observations and/or the deployment of **MonitoringFacility(s)** and **MonitoringNetwork(s)** on the field. Usually, an Environmental Monitoring Programme has a long-term perspective over at least a few years.

The following attributes further characterize the **MonitoringProgramme**:

- **id**: Identifier;
- **name**: Plain text denotation of the **MonitoringProgramme**;
- **additionalDescription**: Plain text description of additional information not fitting in other attributes;
- **mediaMonitored**: Monitored environmental medium;
- **legalBackground**: The legal context, in which the management and regulation of the **MonitoringProgramme** is defined;
- **responsibleParty**: Responsible party for the **MonitoringProgramme**;
- **geometry**: Geometry associated to the **MonitoringProgramme**. For mobile facilities the geometry represents the area the facility is expected to measure in;
- **onlineResource**: A link to an external document providing further information on the **MonitoringProgramme**;
- **purpose**: Reason for which the **MonitoringProgramme** has been generated.

In addition, the following associations provide contextual information:

- **observingCapability**: A link pointing to the explicit capability of an **MonitoringProgramme**. This provides a clean link between the observed property, the procedure used as well as the location of the measurement;
- **broader**: A link pointing to a broader **MonitoringProgramme** (a higher level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;

- **narrower:** A link pointing to narrower **MonitoringProgramme(s)** (a lower level in a hierarchical structure). The association has additional properties as defined in the association class **Hierarchy**;
- **supersedes:** In a genealogy, the **MonitoringProgramme(s)** that has(have) been deactivated/replaced by another one;
- **supersededBy:** In a genealogy, the newly active **MonitoringProgramme(s)** that replaces(replace) the superseded one;
- **triggers:** **MonitoringActivity(s)** triggered by the **MonitoringProgramme**.

7.3.5 MonitoringActivity

Specific set of **MonitoringFacility(s)** and **MonitoringNetwork(s)** used for a given domain in a coherent and concise timeframe, area and purpose. Usually, the information collected is treated as one time step in a long-term monitoring programme. It is a concrete realization of a given **MonitoringProgramme**.

For example, a vessel could be equipped with a collection of **MonitoringFacility(s)** for a given campaign (= **MonitoringActivity**) fulfilling an **MonitoringProgramme** requirements. Then, after a given period this exact same vessel could be equipped with another set of **EnvironmentalMonitoringFacilities** for another campaign fulfilling another **MonitoringProgramme** requirements.

The following attributes further characterize the **MonitoringActivity**:

- **id:** Identifier;
- **activityTime:** Lifespan of the **MonitoringActivity**;
- **activityConditions:** Textual description of the **MonitoringActivity**;
- **boundingBox:** Bounding box in which the **MonitoringActivity** takes place.
Example: If a research vessel has several monitoring activities (**MonitoringActivity**) one wants to know where he cruised for each of those (**EnvironmentalMonitoringActivity**);
- **responsibleParty:** Responsible party for the **MonitoringActivity**;
- **onlineResource:** A link to an external document providing further information on the **MonitoringActivity**.

In addition, the following associations provide contextual information:

- **uses:** Specific set of **MonitoringFacility(s)** and **MonitoringNetwork(s)** involved in an **MonitoringActivity**;
- **setUpFor:** **MonitoringProgramme(s)** for which the **MonitoringActivity** is set up.

	An Observation shall be defined as an act carried out by an Observer to determine the value of an ObservableProperty of an object (featureOfInterest) by using an ObservingProcedure ; the value is provided as the result .
Req 2	/req/req-class-Observation/type-sem Information providing further detail on the type of Observations being described. If information on the type of Observation is provided, the property observationType:ObservationType shall be used.
Req 3	/req/req-class-Observation/observationType-sem If information on the type of Observation is provided, the constraints defined in the referencedodelist shall be used.
Req 4	/req/req-class-Observation/parameter-sem Arbitrary event-specific parameter relevant to the ObservationCharacteristics. If additional parameter information is provided, the property parameter:NamedValue shall be used.
Req 5	/req/req-class-Observation/parameterName-card The name attribute of a parameter NamedValue shall be unique within an Observation.
Req 6	/req/req-class-Observation/resultQuality-sem Information pertaining to the data quality of the result. If additional data quality information is provided, the property resultQuality:Any shall be used.
Req 7	/req/req-class-Observation/phenomenonTime-sem The time for which the result applies to the characteristic of the FeatureOfInterest being observed. If the phenomenonTime is described, this shall be provided by the attribute phenomenonTime:TM_Object
Req 8	/req/req-class-Observation/phenomenonTime-card An Observation shall have exactly 1 phenomenonTime.
Req 9	/req/req-class-Observation/resultTime-sem The instant of time when the result of the Observation became available.

	If the resultTime is described, this shall be provided by the attribute resultTime:TM_Object
Req 10	/req/req-class-Observation/resultTime-card An Observation shall have exactly 1 resultTime .
Req 11	/req/req-class-Observation/resultTime-type If the result time of the Observation is provided, the resultTime attribute shall be of type TM_Instant .
Req 12	/req/req-class-Observation/validTime-sem The time interval during which the result is assumed to be applicable for use. If validTime(s) are described they shall be provided by the attribute validTime:TM_Period
Req 13	/req/req-class-Observation/pFoI-sem The entity that is directly of interest in the act of observing. If a reference to the entity being directly observed is provided, the association with the role proximateFeatureOfInterest shall be used. This association is a specialization of the featureOfInterest role.
Req 14	/req/req-class-Observation/uFoI-sem The entity that is ultimately of interest in the act of observing. If a reference to the entity ultimately being observed is provided, the association with the role ultimateFeatureOfInterest shall be used. This association is a specialization of the featureOfInterest role.
Req 15	/req/req-class-Observation/featureOfInterest-con At least one proximateFeatureOfInterest or ultimateFeatureOfInterest shall be given.
Req 16	/req/req-class-Observation/observedProperty-sem The ObservableProperty that is the subject of the Observation . If a reference to an ObservableProperty is provided, the association with the role observedProperty shall be used.
Req 17	/req/req-class-Observation/observedProperty-card An Observation shall have exactly 1 observedProperty .
Req 18	/req/req-class-Observation/observedProperty-con

	The ObservableProperty referenced by observedProperty shall correspond to a characteristic associated with the featureOfInterest
Req 19	/req/req-class-Observation/result-sem The result of the Observation. If a reference to a result is provided, the association Range with the role result shall be used.
Req 20	/req/req-class-Observation/result-card An Observation shall have exactly 1 result.
Req 21	/req/req-class-Observation/result-con The type of the result provided by the result association shall be suitable for the associated ObservableProperty
Req 22	/req/req-class-Observation/observingProcedure-sem The ObservingProcedure used by the Observation to determine the value of the ObservableProperty provided by the result. If a reference to an ObservingProcedure is provided, the association with the role procedure shall be used.
Req 23	/req/req-class-Observation/observingProcedure-card An Observation shall have exactly 1 observingProcedure.
Req 24	/req/req-class-Observation/observingProcedure-con The ObservingProcedure referenced by procedure shall be suitable for the associated ObservableProperty.
Req 25	/req/req-class-Observation/observer-sem An Observer that is involved in the creation of this Observation. If a reference to an Observer is provided, the association with the role observer shall be used.
Req 26	/req/req-class-Observation/host-sem A Host that is involved in the creation of this Observation If a reference to a Host is provided, the association with the role host shall be used.
Req 27	/req/req-class-Observation/observerhost-con At least one Observer or Host shall be provided
Req 28	/req/req-class-Observation/uom

	The Observation shall provide a unit of measure (UoM) if the result is measurable. If the UoM is not contained in the result, it shall be provided in the context of the Observation ; the provision modality is to be defined by communities.
Req 29	/req/obs-cpt/gen/relatedObservation-sem An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 30	/req/req-class-Observation/metadata If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.
Rec 1	/rec/req-class-ObservationCharacteristics/parameter-procedure Parameter should not be used instead of the procedure to describe the steps performed in order to determine the value of the ObservableProperty.
Rec 2	/rec/req-class-ObservationCharacteristics/parameter-redundant Parameter should not be utilized to provide information already contained in the model by existing attributes or associations.
Rec 3	/rec/req-class-ObservationCharacteristics/uFoI In the case where ultimate and proximate features-of-interest are the same object, the association should be provided using the ultimateFeatureOfInterest association role.
Rec 4	/rec/req-class-Observation/phenomenonTimeResult-con If the observedProperty of an Observation is ‘occurrence time’ then the result should be the same as the phenomenonTime.
Rec 5	/rec/req-class-Observation/uom-con The unit of measure should be suitable for the associated ObservableProperty and ObservingProcedure

8.3 ObservableProperty

An **ObservableProperty** shall be defined as a quality (property, characteristic) of the feature-of-interest that can be observed.

Requirements Class
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservableProperty

Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-ObservableProperty/ObservableProperty-sem An ObservableProperty shall be defined as a quality (property, characteristic) of the feature-of-interest that can be observed.
Req 2	/req/req-class-ObservableProperty/observer-sem An Observer capable of observing this ObservableProperty. If a reference to the Observer is provided, the association with the role observer shall be used.
Req 3	/req/req-class-ObservableProperty/relatedObservation-sem An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 4	/req/req-class-ObservableProperty/metadata If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

8.4 ObservingProcedure

An **ObservingProcedure** shall be defined as the description of steps performed in order to determine the value of an **observableProperty** by an **Observer**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservingProcedure	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-ObservingProcedure/ObservingProcedure-sem An ObservingProcedure shall be defined as the description of steps performed in order to determine the value of an observableProperty by an Observer.
Req 2	/req/req-class-ObservingProcedure/observer-sem An Observer capable of performing this ObservingProcedure.

	If a reference to an Observer is provided, the association with the role observer shall be used.
Req 3	/req/req-class-ObservingProcedure/relatedObservation-sem An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 4	/req/req-class-ObservingProcedure/metadata If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

8.5 Observer

An **Observer** shall be defined as an identifiable entity that can generate **Observations** pertaining to an **observableProperty** by implementing an **ObservingProcedure**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Observer	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-Observer/Observer-sem An Observer shall be defined as an identifiable entity that can generate Observations pertaining to an observableProperty by implementing an ObservingProcedure.
Req 2	/req/req-class-Observer/observableProperty-sem An ObservableProperty that this Observer can observe. If a reference to ObservableProperty(s) is provided, the association with the role observableProperty shall be used.
Req 3	/req/req-class-Observer/observingProcedure-sem An ObservingProcedure that this Observer can perform. If a reference to ObservingProcedure(s) is provided, the association with the role observingProcedure shall be used.
Req 4	/req/req-class-Observer/deployment-sem A Deployment to which this Observer is either physically or organizationally attached.

	If a reference to Deployment(s) is provided, the association with the role deployment shall be used.
Req 5	/req/req-class-Observer/location-sem Location information pertaining to a feature. If location information is provided, the attribute location:Geometry shall be used.
Req 6	/req/req-class-Observer/relatedObservation-sem An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 6	/req/req-class-Observer/metadata If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

8.6 Host

A **Host** shall be defined as a grouping of **Observers** for a specific reason.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Host	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-Host/Host-sem A Host shall be defined as a grouping of Observers for a specific reason.
Req 2	/req/req-class-Host/deployment-sem A Deployment at this Host . If a reference to a Deployment is provided, the association with the role host shall be used.
Req 3	/req/req-class-Host/relatedHost-sem A Host the Host is related to. If a reference to a related Host is provided, the association with role relatedHost shall be used. The context:GenericName qualifier of this

	association may be used to provide further information as to the nature of the relation.
Req 4	/req/req-class-Host/location-sem Location information pertaining to a feature. If location information is provided, the attribute location:Geometry shall be used.
Req 5	/req/req-class-Host/relatedObservation-sem An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 6	/req/req-class-Host/metadata If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

8.7 Deployment

A **Deployment** shall be defined as information on the assignment of an **Observer** to a **Host**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Deployment	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-Deployment/Deployment-sem A Deployment shall be defined as information on the assignment of an Observer to a Host .
Req 2	/req/req-class-Deployment/observer-sem The Observer associated with this Deployment . If a reference to an Observer is provided, the association with the role observer shall be used.
Req 3	/req/req-class-Deployment/host-sem The Host to which this Deployment pertains.

	If a reference to a Host is provided, the association with the role host shall be used
Req 4	/req/req-class-Deployment/deploymentReason-sem A human readable description of the reason for the Deployment . If the reason for the Deployment is provided, the property deploymentReason:CharacterString shall be used.
Req 5	/req/req-class-Deployment/deploymentTime-sem The time that the Deployment pertains to. If the time of the Deployment is provided, property deploymentTime:TM Period shall be used.
Req 6	/req/req-class-Deployment/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

8.8 ObservationCharacteristics

An **ObservationCharacteristics** shall be defined as a set of common characteristics used for describing an **Observation** or a collection of Observations.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCharacteristics	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-ObservationCharacteristics/ObservationCharacteristics-sem An ObservationCharacteristics shall be defined as a set of common characteristics used for describing an Observation or a collection of Observations.
Req 2	/req/req-class-ObservationCharacteristics/type-sem Information providing further detail on the type of Observations being described by the ObservationCharacteristics . If information on the type of Observation is provided, the property observationType:ObservationType shall be used.
Req 3	/req/req-class-ObservationCharacteristics/parameter-sem

	Arbitrary event-specific parameter relevant to the ObservationCharacteristics. If additional parameter information is provided, the property parameter:NamedValue shall be used.
Req 4	/req/req-class-ObservationCharacteristics/resultQuality-sem Information pertaining to the data quality of the result. If additional data quality information is provided, the property resultQuality:Any shall be used.
Req 5	/req/req-class-ObservationCharacteristics/phenomenonTime-sem The time for which the result applies to the characteristic of the FeatureOfInterest being observed. If the phenomenonTime is described, this shall be provided by the attribute phenomenonTime:TM_Object
Req 6	/req/req-class-ObservationCharacteristics/resultTime-sem The instant of time when the result of the Observation became available. If the resultTime is described, this shall be provided by the attribute resultTime:TM_Object
Req 7	/req/req-class-ObservationCharacteristics/validTime-sem The time interval during which the result is assumed to be applicable for use. If validTime(s) are described they shall be provided by the attribute validTime:TM_Period
Req 8	/req/req-class-ObservationCharacteristics/featureOfInterest-sem The subject of the Observation. The reference(s) to featureOfInterest(s) shall be provided using the association Domain with the role featureOfInterest.
Req 9	/req/req-class-ObservationCharacteristics/pFoI-sem The entity that is directly of interest in the act of observing. If a reference to the entity being directly observed is provided, the association with the role proximateFeatureOfInterest shall be used. This association is a specialization of the featureOfInterest role.
Req 10	/req/req-class-ObservationCharacteristics/uFoI-sem

	The entity that is ultimately of interest in the act of observing. If a reference to the entity ultimately being observed is provided, the association with the role ultimateFeatureOfInterest shall be used. This association is a specialization of the featureOfInterest role.
Req 11	/req/obs-basic/ObservationCharacteristics/collection-sem An ObservationCollection that is described by these ObservationCharacteristics. If a reference to an ObservationCollection <i>from the</i> ObservationCharacteristics is provided, the association with the role collection shall be used.
Req 12	/req/req-class-ObservationCharacteristics/observedProperty-sem The ObservableProperty that is the subject of the Observation. If a reference to an ObservableProperty is provided, the association with the role observedProperty shall be used.
Req 13	/req/req-class-ObservationCharacteristics/result-sem The result of the Observation. If a reference to a result is provided, the association Range with the role result shall be used.
Req 14	/req/req-class-ObservationCharacteristics/observingProcedure-sem The ObservingProcedure used by the Observation to determine the value of the ObservableProperty provided by the result. If a reference to an ObservingProcedure is provided, the association with the role procedures shall be used.
Req 15	/req/req-class-ObservationCharacteristics/observer-sem An Observer that is involved in the creation of this Observation. If a reference to an Observer is provided, the association with the role observer shall be used.
Req 16	/req/req-class-ObservationCharacteristics/host-sem A Host that is involved in the creation of this Observation If a reference to a Host is provided, the association with the role host shall be used.
Req 17	/req/req-class-ObservationCharacteristics/relatedObservation-sem

	An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 18	/req/req-class-ObservationCharacteristics/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.
Rec 1	/rec/req-class-ObservationCharacteristics/parameter-procedure Parameter should not be used instead of the procedure to describe the steps performed in order to determine the value of the ObservableProperty.
Rec 2	/rec/req-class-ObservationCharacteristics/parameter-redundant Parameter should not be utilized to provide information already contained in the model by existing attributes or associations.
Rec 3	/rec/req-class-ObservationCharacteristics/uFoI In the case where ultimate and proximate features-of-interest are the same object, the association should be provided using the ultimateFeatureOfInterest association role.

8.9 ObservationCollection

An **ObservationCollection** shall be defined as a collection of similar **Observations**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCollection	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-ObservationCollection/ObservationCollection-sem An ObservationCollection shall be defined as a collection of similar Observations.
Req 2	/req/req-class-ObservationCollection/collectionType-sem Information on the type of the ObservationCollection. If information on the collection type is provided, the attribute collectionType:ObservationCollectionType shall be used.
Req 3	/req/req-class-ObservationCollection/collectionType-con

	If the collectionType is provided, property values of the associated Observation and ObservationCharacteristics instances shall comply with the constraints defined for this collectionType value.
Req 4	/req/req-class-ObservationCollection/member-sem An Observation that is part of this ObservationCollection. If a reference to a member Observation is provided, the association with the role member shall be used.
Req 5	/req/req-class-ObservationCollection/memberCharacteristics-sem Information on ObservationCharacteristics of Observations contained within the ObservationCollection. If a reference to ObservationCharacteristics pertaining to the collection members is provided, the association with the role memberCharacteristics shall be used.
Req 6	/req/req-class-ObservationCollection/relatedCollection-sem A ObservationCollection the ObservationCollection is related to. If a reference to a related ObservationCollection is provided, the association with role relatedCollection shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 7	/req/req-class-ObservationCollection/relatedObservation-sem An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.

8.10 NamedValue

The class **NamedValue** provides for a generic soft-typed parameter value.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-NamedValue	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class

Req 1	/req/req-class-NamedValue/NamedValue-sem The class NamedValue provides for a generic soft-typed parameter value.
Req 2	/req/req-class-NamedValue/name-sem The attribute name:GenericName shall indicate the meaning of the named value.
Req 3	/req/req-class-NamedValue/valu-sem The attribute value:Any shall provide the value.

8.11 Codelist ObservationType

An empty extension-point for providing various classification schemes for **Observations**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationType	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-ObservationType/ObservationType-sem An empty extension-point for providing various classification schemes for Observations . If Observation classification schemes are used in the implementing application schemas, a concrete realization shall be created for the application.

8.12 Codelist ObservationTypeByResultType

A codelist for classifying Observations by their result type

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationTypeByResultType	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-ObservationTypeByResultType/ObservationTypeByResultType-sem The following entries shall be provided: • measurement: the result is of type Measure;

	• category-observation: the result is of type ScopedName; • truth-observation: result is a truth value; • count-observation: the result is of type Integer; • temporal-observation: the result is of type TM_Object; • geometry-observation: the result is of type Geometry; • complex-observation: the result is of type Record; • discrete-coverage-observation: result is a coverage that returns the same feature attribute values for every direct position within any single spatial object, temporal object, or spatiotemporal object in its domain; • discrete-point-coverage: result is a coverage that has a domain composed of points; • timeseries-observation: the result is a timeseries (a sequence of data values which are ordered in time).
Req 2	/req/req-class-ObservationTypeByResultType/ObservationTypeByResultType-sem The following constraints shall be applied to the value of the result association of the Observation based on the codelist value used: • If the value "measurement" is used, the value of the result shall be of type Measure; • If the value "category-observation" is used the value of the result shall be of type ScopedName; • If the value "truth-observation" is used, the value of result shall be a truth value; • If the value "count-observation" is used, the value of the result shall be of type Integer; • If the value "temporal-observation" is used, the value of the result shall be of type TM_Object; • If the value "geometry-observation" is used, the value of the result shall be of type Geometry;

	If the value "complex-observation" is used, the value of the result shall be of type Record.
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8.13 Codelist ObservationCollectionType

A codelist for classifying ObservationCollections.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCollectionType	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-ObservationCollectionType/ObservationCollectionType-sem The following entries shall be provided: homogeneous: all observations contained are of a similar nature; summarizing: a wider grab-bag type of collection.
Req 2	/req/req-class-ObservationCollectionType/homogeneous-con If collectionType in the ObservationCollection is specified as “homogeneous” from this Codelist, the following constraints apply to the associated ObservationCharacteristics and all Observation instances referenced via the member association. If a property value is provided within the ObservationCharacteristics, this value applies to all Observations contained in the ObservationCollection: - property not provided – values may be provided by the Observations but is not provided at this level - property provided but with no content – no Observation within the collection provides this property - property = value – this value applies to all Observations within the collection property = value set/range – this value set/range applies to all Observations within the collection
Req 3	/req/req-class-ObservationCollectionType/summarizing-con If collectionType in the ObservationCollection is specified as “summarizing” from this Codelist, the following constraints apply to the

	associated ObservationCharacteristics and all Observation instances referenced via the member association. If multiple values for a property are available in the contained Observations, all values for this attribute (or the range of values contained in all Observations) are provided in the ObservationCharacteristics. A property may also be empty in the ObservationCharacteristics – in this case any value can be provided for this attribute within the contained Observations: • property not provided – values may be provided by the Observations but a summary is not provided at this level; • property provided but with no content – no Observation within the collection provides this property; • property = value – this value applies to all Observations within the collection; • property = value set/range – all Observations provide a value within this set/range.
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9. Logical Sampling Schema

9.1 General

The Core Logical Sampling schema is described as a class diagram in Figure 4.

In Figure 5 we see the additional classes SampleCollection as well as the specialized Sample types StatisticalSample, MaterialSample and SpatialSample.

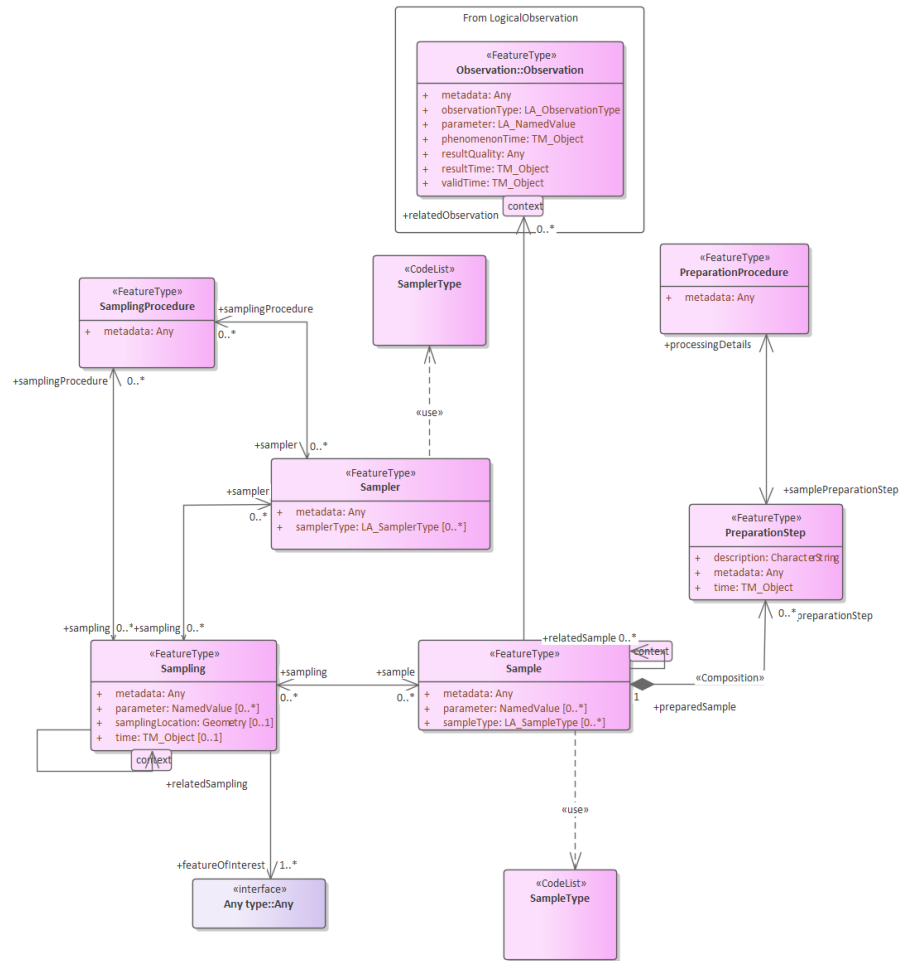


Figure 4: Core Sampling Schema

	If Sampling(s) are described they shall be referred to using the association with the role sampling.
Req 3	/req/req-class-Sample/preparationStep-sem The PreparationStep(s) applied to prepare the Sample. If PreparationSteps are described they shall be referred to using the association with the role preparationStep.
Req 4	/req/req-class-Sample/sampledFeature-sem The sampledFeature is the feature the Sample is intended to be representative of. References to the sampled feature shall be provided using the association with the role sampledFeature.
Req 5	/req/req-class-Sample/sampleType-sem The type of Sample according to a community agreed typology. If information on the type of Sample is provided, the attribute sampleType:SampleType shall be used.
Req 6	/req/req-class-Sample/parameter-sem Arbitrary event-specific parameter relevant to the Sample. If additional parameter information is provided, the property parameter:NamedValue shall be used.
Req 7	/req/req-class-Sample/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.
Req 8	/req/req-class-Sample/relatedSample-sem A Sample the Sample is related to. If a reference to a related Sample is provided, the association with role relatedSample shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 9	/req/req-class-Sample/relatedObservation-sem An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName

	qualifier of this association may be used to provide further information as to the nature of the relation.
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9.3 SpatialSample

A **SpatialSample** shall be defined as a geospatial **Sample**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SpatialSample	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-SpatialSample/Sample-sem A Sample shall be defined as an object that is representative of a concept, real-world object or phenomenon.
Req 2	/req/req-class-SpatialSample/sampling-sem The Sampling the Sample is the result of. If Sampling(s) are described they shall be referred to using the association with the role sampling.
Req 3	/req/req-class-SpatialSample/preparationStep-sem The PreparationStep(s) applied to prepare the Sample. If PreparationSteps are described they shall be referred to using the association with the role preparationStep.
Req 4	/req/req-class-SpatialSample/sampledFeature-sem The sampledFeature is the feature the Sample is intended to be representative of. References to the sampled feature shall be provided using the association with the role sampledFeature.
Req 5	/req/req-class-SpatialSample/sampleType-sem The type of Sample according to a community agreed typology. If information on the type of Sample is provided, the attribute sampleType:SampleType shall be used.
Req 6	/req/req-class-SpatialSample/parameter-sem Arbitrary event-specific parameter relevant to the Sample.

	If additional parameter information is provided, the property parameter:NamedValue shall be used.
Req 7	/req/req-class-SpatialSample/SpatialSample-sem A SpatialSample shall be defined as a geospatial Sample.
Req 8	/req/req-class-SpatialSample/shape-sem The shape is the Geometry of the SpatialSample. If location information pertaining to the SpatialSample is provided, the attribute shape:Geometry shall be used.
Req 9	/req/req-class-SpatialSample/horizontalPositionalAccuracy-sem The positional accuracy of the horizontal component of the Geometry of the SpatialSample. If horizontal positional accuracy information pertaining to the SpatialSample is provided, the attribute horizontalPositionalAccuracy:Any shall be used.
Req 10	/req/req-class-SpatialSample/verticalPositionalAccuracy-sem The positional accuracy of the vertical component of the Geometry of the SpatialSample. If horizontal positional accuracy information pertaining to the SpatialSample is provided, the attribute verticalPositionalAccuracy:Any shall be used.
Req 11	/req/req-class-SpatialSample/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.
Req 12	/req/req-class-SpatialSample/relatedSample-sem A Sample the Sample is related to. If a reference to a related Sample is provided, the association with role relatedSample shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 13	/req/req-class-SpatialSample/relatedObservation-sem An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName

	qualifier of this association may be used to provide further information as to the nature of the relation.
--	--

9.4 MaterialSample

A **MaterialSample** shall be defined as a physical, tangible **Sample**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MaterialSample	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MaterialSample/Sample-sem A Sample shall be defined as an object that is representative of a concept, real-world object or phenomenon.
Req 2	/req/req-class-MaterialSample/sampling-sem The Sampling the Sample is the result of. If Sampling(s) are described they shall be referred to using the association with the role sampling.
Req 3	/req/req-class-MaterialSample/preparationStep-sem The PreparationStep(s) applied to prepare the Sample. If PreparationSteps are described they shall be referred to using the association with the role preparationStep.
Req 4	/req/req-class-MaterialSample/sampledFeature-sem The sampledFeature is the feature the Sample is intended to be representative of. References to the sampled feature shall be provided using the association with the role sampledFeature.
Req 5	/req/req-class-MaterialSample/sampleType-sem The type of Sample according to a community agreed typology. If information on the type of Sample is provided, the attribute sampleType:SampleType shall be used.
Req 6	/req/req-class-MaterialSample/parameter-sem Arbitrary event-specific parameter relevant to the Sample.

	If additional parameter information is provided, the property parameter:NamedValue shall be used.
Req 7	/req/req-class-MaterialSample/MaterialSample-sem A MaterialSample shall be defined as a physical, tangible Sample.
Req 8	/req/req-class-MaterialSample/size-sem The size describes a physical extent of the MaterialSample. If size information pertaining to the MaterialSample is provided, the attribute size:PhysicalDimension shall be used.
Req 9	/req/req-class-MaterialSample/storageLocation-sem The storageLocation is the location of a MaterialSample. If information pertaining to the storage location of the MaterialSample is provided, the attribute storageLocation:NamedLocation shall be used.
Req 10	/req/req-class-MaterialSample/sourceLocation-sem The sourceLocation is the location from where the MaterialSample was obtained. If information pertaining to the source location of the MaterialSample is provided, the attribute sourceLocation:Geometry shall be used.
Req 11	/req/req-class-MaterialSample/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.
Req 12	/req/req-class-MaterialSample/relatedSample-sem A Sample the Sample is related to. If a reference to a related Sample is provided, the association with role relatedSample shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 13	/req/req-class-MaterialSample/relatedObservation-sem An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.

9.5 StatisticalSample

A **StatisticalSample** shall be defined as a statistical subset of a feature-of-interest, defined for the purpose of creating **Observation(s)**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-StatisticalSample	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-StatisticalSample/Sample-sem A Sample shall be defined as an object that is representative of a concept, real-world object or phenomenon.
Req 2	/req/req-class-StatisticalSample/sampling-sem The Sampling the Sample is the result of. If Sampling(s) are described they shall be referred to using the association with the role sampling.
Req 3	/req/req-class-StatisticalSample/preparationStep-sem The PreparationStep(s) applied to prepare the Sample. If PreparationSteps are described they shall be referred to using the association with the role preparationStep.
Req 4	/req/req-class-StatisticalSample/sampledFeature-sem The sampledFeature is the feature the Sample is intended to be representative of. References to the sampled feature shall be provided using the association with the role sampledFeature.
Req 5	/req/req-class-StatisticalSample/sampleType-sem The type of Sample according to a community agreed typology. If information on the type of Sample is provided, the attribute sampleType:SampleType shall be used.
Req 6	/req/req-class-StatisticalSample/parameter-sem Arbitrary event-specific parameter relevant to the Sample. If additional parameter information is provided, the property parameter:NamedValue shall be used.
Req 7	/req/req-class-StatisticalSample/StatisticalSample-sem

	A StatisticalSample shall be defined as a statistical subset of a feature-of-interest, defined for the purpose of creating Observation(s) .
Req 8	/req/req-class-StatisticalSample/classification-sem The classification describes a criterion by which the subset was defined. If information pertaining to the subsetting criteria by which a StatisticalSample has been defined is provided, the attribute classification:StatisticalClassification shall be used.
Req 9	/req/req-class-StatisticalSample/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.
Req 10	/req/req-class-StatisticalSample/relatedSample-sem A Sample the Sample is related to. If a reference to a related Sample is provided, the association with role relatedSample shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
Req 11	/req/req-class-StatisticalSample/relatedObservation-sem An Observation the object is related to. If a reference to a related Observation is provided, the association with role relatedObservation shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.

9.6 Sampling

A **Sampling** shall be defined as an act applying a **SamplingProcedure** to create or transform one or more **Sample(s)**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sampling	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-Sampling/Sampling-sem A Sampling shall be defined as an act applying a SamplingProcedure to create or transform one or more Sample(s) .
Req 2	/req/req-class-Sampling/sample-sem

	The Sample generated by the Sampling. If Samples are described they shall be referred to using the association with the role sample.
Req 3	/req/req-class-Sampling/featureOfInterest-sem A feature-of-interest shall be defined as the concept, real-world object or phenomenon (feature-of-interest) the Sample(s) of the Sampling represent.
Req 4	/req/req-class-Sampling/featureOfInterest-card Reference to the feature-of-interest shall be done using the association with the role featureOfInterest.
Req 5	/req/req-class-Sampling/sampler-sem The Sampler that performed the Sampling. If Sampler(s) are described they shall be referred to using the association with the role sampler.
Req 6	/req/req-class-Sampling/samplingProcedure-sem The SamplingProcedure used by the Sampling. If SamplingProcedures are described they shall be referred to using the association with the role samplingProcedure.
Req 7	/req/req-class-Sampling/samplingLocation-sem If location information pertaining to the Sampling is provided, the attribute samplingLocation:Geometry shall be used.
Req 8	/req/req-class-Sampling/time-sem If information on the time of the Sampling is provided, the attribute time:TM_Object shall be used.
Req 9	/req/req-class-Sampling/parameter-sem Arbitrary event-specific parameter relevant to the Sampling. If additional parameter information is provided, the property parameter:NamedValue shall be used.
Req 10	/req/req-class-Sampling/relatedSampling-sem Related Sampling(s). If a reference to a related Sampling is provided, the association with role relatedSampling shall be used. The context:GenericName qualifier of

	this association may be used to provide further information as to the nature of the relation.
Req 11	/req/req-class-Sampling/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

9.7 Sampler

A **Sampler** shall be defined as a device or entity (including humans) that is used by, or implements, a **SamplingProcedure** to create or transform one or more **Sample(s)**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sampler	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-Sampler/Sampler-sem A Sampler shall be defined as a device or entity (including humans) that is used by, or implements, a SamplingProcedure to create or transform one or more Sample(s) .
Req 2	/req/req-class-Sampler/sampling-sem The Sampling act performed by the Sampler . If Sampling(s) are described they shall be referred to using the association with the role sampling .
Req 3	/req/req-class-Sampler/implementedProcedure-sem The Procedure implemented by the Sampler . If Procedure(s) are described they shall be referred to using the association with the role implementedProcedure .
Req 4	/req/req-class-Sampler/samplerType-sem The type of Sampler according to a community agreed typology. If information on the type of Sampler is provided, the attribute samplerType:SamplerType shall be used.
Req 5	/req/req-class-Sampler/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

9.8 SamplingProcedure

A **SamplingProcedure** shall be defined as the description of steps performed by a **Sampler** in order to extract a **Sample** from its **sampledFeature** in the frame of a **Sampling**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SamplingProcedure	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-SamplingProcedure/SamplingProcedure-sem A SamplingProcedure shall be defined as the description of steps performed by a Sampler in order to extract a Sample from its sampledFeature in the frame of a Sampling .
Req 2	/req/req-class-SamplingProcedure/sampling-sem If the SamplingProcedure provides information on the Sampling where this procedure has been used, the association with the role sampling shall be used.
Req 3	/req/req-class-SamplingProcedure/sampler-sem If the SamplingProcedure provides information on the Sampler that implements this procedure, the association with the role sampler shall be used.
Req 4	/req/req-class-SamplingProcedure/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

9.9 PreparationStep

A **PreparationStep** shall be defined as an individual step pertaining to a **PreparationProcedure**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PreparationStep	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-PreparationStep/PreparationStep-sem A PreparationStep shall be defined as an individual step pertaining to a PreparationProcedure .

Req 2	/req/req-class-PreparationStep/processingDetails-sem A PreparationProcedure step performed on the Sample the PreparationStep pertains to. If PreparationProcedure(s) are described they shall be referred to using the association with the role processingDetails.
Req 3	/req/req-class-PreparationStep/preparedSample-sem The Sample on which the PreparationProcedure is performed. The Sample shall be referred to using the association with the role preparedSample.
Req 4	/req/req-class-PreparationStep/description-sem Description of the preparationStep. If a description pertaining to the preparationStep is provided, the attribute description:CharacterString shall be used.
Req 5	/req/req-class-PreparationStep/time-sem Time of the preparationStep. If information on the time of the preparationStep of the Sampling is provided, the attribute time:TM Object shall be used.
Req 6	/req/req-class-PreparationStep/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

9.10 PreparationProcedure

A **PreparationProcedure** shall be defined as the description of preparation steps performed on a **Sample**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PreparationProcedure	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-PreparationProcedure/PreparationProcedure-sem A PreparationProcedure shall be defined as the description of preparation steps performed on a Sample.
Req 2	/req/req-class-PreparationProcedure/samplePreparationStep-sem

	If the PreparingProcedure provides information on the PreparationStep where this procedure has been used, the association with the role samplePreparationStep shall be used.
Req 3	/req/req-class-PreparationProcedure/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

9.11 SampleCollection

A **SampleCollection** shall be defined as a collection of **Samples**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleCollection	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-SampleCollection/SampleCollection-sem A SampleCollection shall be defined as a collection of Samples.
Req 2	/req/req-class-SampleCollection/member-sem A Sample that is part of this SampleCollection. If the SampleCollection has members, the association with the role member shall be used.
Req 3	/req/req-class-SampleCollection/relatedCollection-sem A SampleCollection the SampleCollection is related to. If a reference to a related SampleCollection is provided, the association with role relatedCollection shall be used. The context:GenericName qualifier of this association may be used to provide further information as to the nature of the relation.
	/req/req-class-SampleCollection/metadata-sem If descriptive metadata is provided, the association role metadata shall link to descriptive metadata as commonly understood by communities.

9.12 PhysicalDimension

A **PhysicalDimension** shall be defined as a dataType for the provision of various size quantities.

Requirements Class
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PhysicalDimension

Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-PhysicalDimension/PhysicalDimension-sem A PhysicalDimension shall be defined as a dataType for the provision of various size quantities.
Req 2	/req/req-class-PhysicalDimension/dimension-sem The name of the PhysicalDimension about which a value is provided. The identifier of the physical dimension shall be provided in the attribute dimension:URI .
Req 3	/req/req-class-PhysicalDimension/value-sem The value of the PhysicalDimension . The measure of the quantity being provided shall be provided in the attribute value:Measure

9.13 NamedLocation

A **NamedLocation** shall be defined as a location identified by its name, address, spatial geometry or a combination of any of these three.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-NamedLocation	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-NamedLocation/NamedLocation-sem A NamedLocation shall be defined as a location identified by its name, address, spatial geometry or a combination of any of these three.
Req 2	/req/req-class-NamedLocation/address-sem An address used for identifying a NamedLocation . If address information is provided, the attribute address:Any shall be used.
Req 3	/req/req-class-NamedLocation/name-sem A name used for identifying a NamedLocation .

	If name information is provided, the attribute name:GenericName shall be used.
Req 4	/req/req-class-NamedLocation/representativeGeometry-sem A geometry used for providing a representative spatial location of a NamedLocation . If geometry is provided, the attribute representativeGeometry:Geometry shall be used.

9.14 StatisticalClassification

A **StatisticalClassification** shall be defined as a dataType for the provision of information on statistical classifications.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-StatisticalClassification	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-StatisticalClassification/StatisticalClassification-sem A StatisticalClassification shall be defined as a dataType for the provision of information on statistical classifications.
Req 2	/req/req-class-StatisticalClassification/concept-sem The concept by which a StatisticalClassification is to be performed. The name of the concept by which the statistical classification is performed shall be provided in the attribute concept:URI .
Req 3	/req/req-class-StatisticalClassification/classification-sem The explicit classification class pertaining to the classification concept described by the StatisticalClassification . The classification class of the StatisticalClassification shall be provided in the attribute classification:URI .

9.15 Codelist SampleType

The code list SampleType can be specialized as required to more precisely define the semantics of sample types.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleType	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-SampleType/SampleType-sem An empty extension-point for providing various classification schemes for Samples. If Sample classification schemes are used in the implementing application schemas, a concrete realization shall be created for the application.

9.16 Codelist SampleTypeByGeometryType

The code list SampleType can be specialized by geometry type

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleTypeByGeometryType	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-SampleTypeByGeometryType/SampleTypeByGeometryType-sem The following entries shall be provided: • point: the provided geometry is of type Point. • curve: the provided geometry is of type Curve. • surface: the provided geometry is of type Surface. • solid: the provided geometry is of type Solid.
Req 2	/req/req-class-SampleTypeByGeometryType/SampleTypeByGeometryType-con The following constraints shall be applied to the value of the result association of the Observation based on the codelist value used: • If value “point” is used, the provided geometry shall be of type Point.

	• If value “curve” is used, the provided geometry shall be of type Curve. • If value “surface” is used, the provided geometry shall be of type Surface. • If value “solid” is used, the provided geometry shall be of type Solid.
--	--

9.17 Codelist **SamplerType**

The code list **SamplerType** can be specialized as required to more precisely define the semantics of sampler types.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SamplerType	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-SamplerType/SamplerType-sem An empty extension-point for providing various classification schemes for Samplers. If Sampler classification schemes are used in the implementing application schemas, a concrete realization shall be created for the application.

10. Logical Monitoring Core

10.1 General

Within the European INSPIRE Environmental Monitoring Facilities specification, one finds the OMS Host concept fully fleshed out, together with relevant contextual information. As these have been proven useful in real world scenarios, they are being provided as an extension in the context of the OMS Logical model. As parts of the full INSPIRE model have a strong European focus, a simplified core model has been derived, reduced to the most essential concepts, described in this section as the core model. In addition to the core model, the next section provides an extended version, including the full richness of the original INSPIRE model.

Please note that as all core classes are modelled as FeatureType, basic attributes such as identifier, name and geometry are already covered by this basic stereotype, they are not provided within this specification.

The core classes of this model, as shown in Figure 6, are:

- **MonitoringFacility:** A georeferenced object directly collecting or processing data about objects whose properties (e.g. physical, chemical, biological or other aspects of environmental conditions) are repeatedly observed or measured.
- **MonitoringNetwork:** Administrative or organisational grouping of EnvironmentalMonitoringFacilities managed the same way for a specific purpose, targeting a specific area.
- **MonitoringActivity:** Specific set of MonitoringFacilities or MonitoringNetworks used for a given domain in a coherent and concise timeframe, area and purpose. Usually, the information collected is treated as one time step in a long term monitoring programme. It is a concrete realisation of a given MonitoringProgramme.
- **MonitoringProgramme:** Framework based on policy relevant documents defining the target of a collection of observations and/or the deployment of MonitoringFacilities or MonitoringNetworks on the field. Usually, a Monitoring Programme has a long-term perspective over at least a few years.

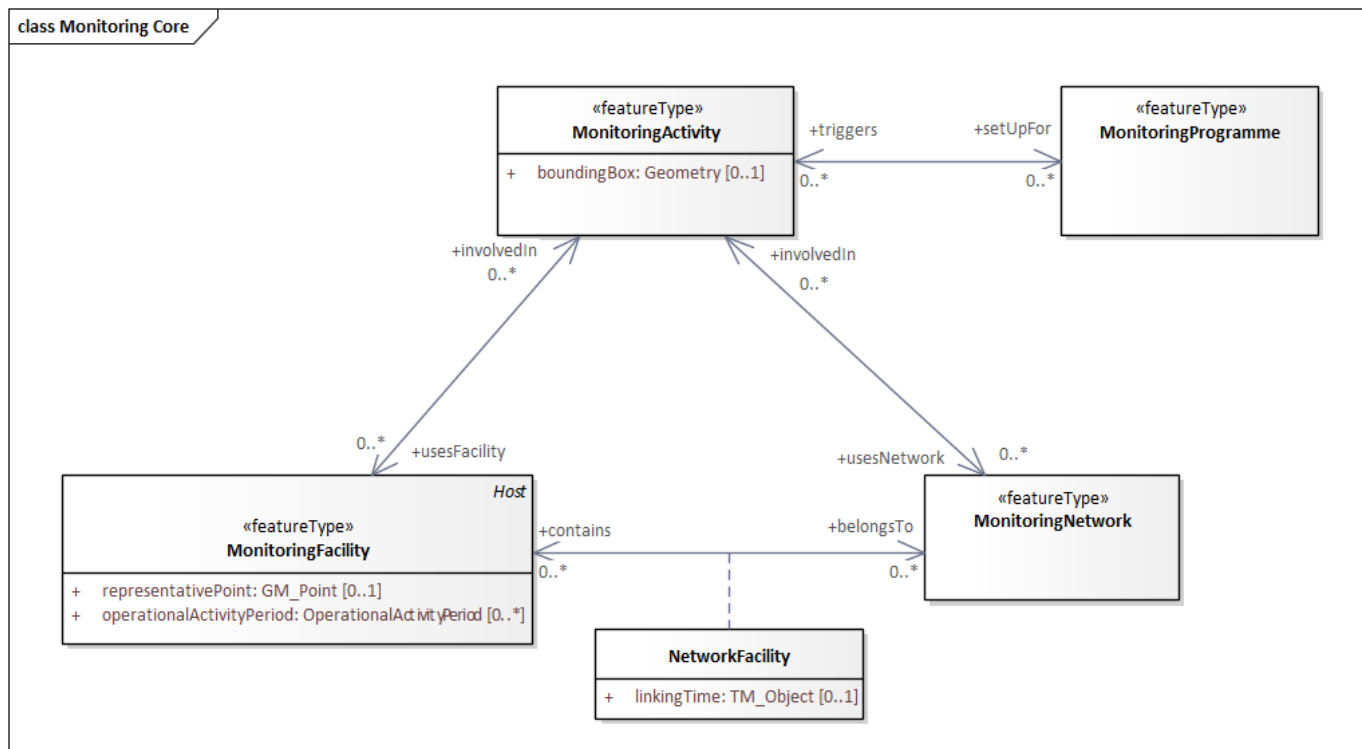


Figure 6: Core Monitoring Schema

In Figure 7 below, the Monitoring Schema is shown with OMS context, illustrating both how MonitoringFacility is derived from OMS:Host as well as how OMS:ObservingCapability is utilized by MonitoringFacility, MonitoringNetwork and MonitoringProgramme.

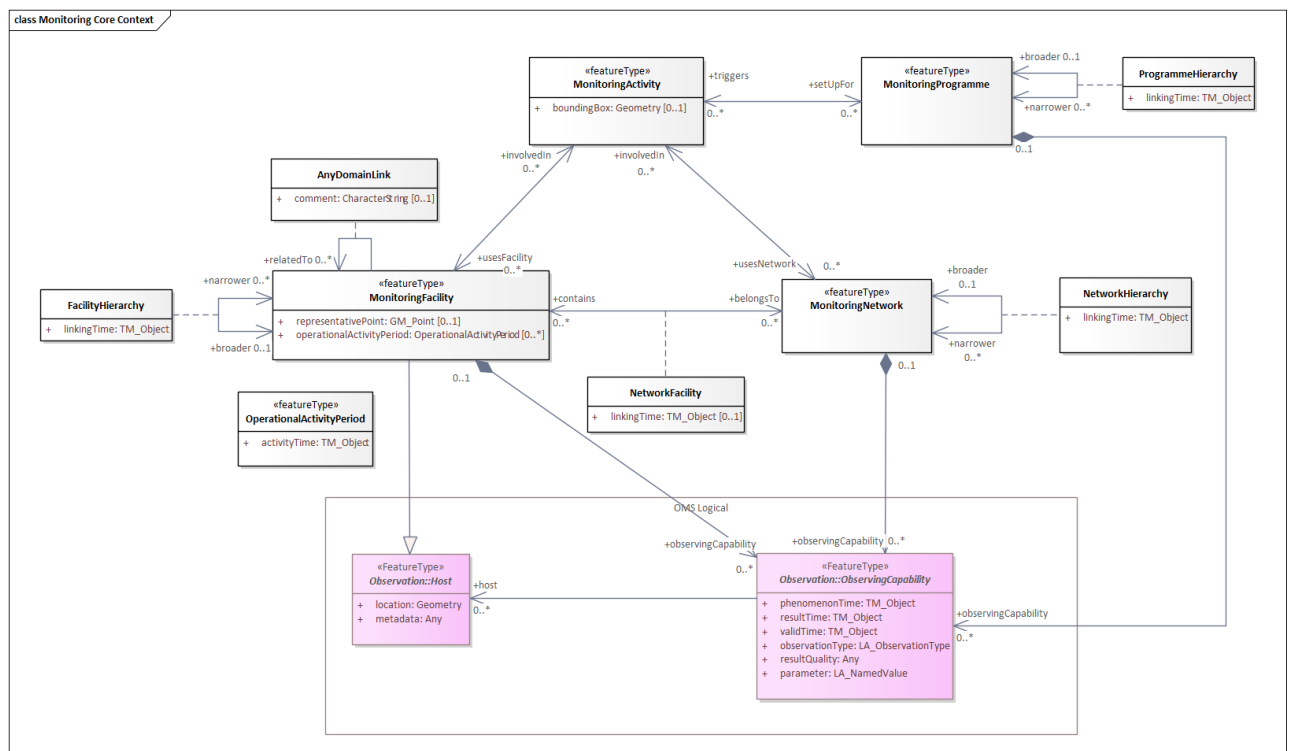


Figure 7: Monitoring Schema - OMS Context

10.2 MonitoringFacility

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacility	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/MonitoringFacility-sem A georeferenced object directly collecting or processing data about objects whose properties (e.g. physical, chemical, biological or other aspects of environmental conditions) are repeatedly observed or measured. An environmental monitoring facility can also host other environmental monitoring facilities.

Req 2	/req/req-class-MonitoringFacility/representativePoint Representative location for the MonitoringFacility .
Req 3	/req/req-class-MonitoringFacility/operationalActivityPeriod Lifespan of the physical object (facility).
Req 4	/req/req-class-MonitoringFacility/observingCapability A link pointing to the explicit capability of an MonitoringFacility . This provides a clean link between the observed property, the procedure used as well as the location of the measurement.
Req 5	/req/req-class-MonitoringFacility/broader A link pointing to a broader MonitoringFacility (a higher level in a hierarchical structure). The association has additional properties as defined in the association class Hierarchy .
Req 6	/req/req-class-MonitoringFacility/narrower A link pointing to narrower MonitoringFacility(s) (a lower level in a hierarchical structure). The association has additional properties as defined in the association class Hierarchy .
Req 7	/req/req-class-MonitoringFacility/belongsTo A link pointing to the MonitoringNetwork(s) this MonitoringFacility pertains to. The association has additional properties as defined in the association class NetworkFacility .
Req 8	/req/req-class-MonitoringFacility/relatedTo Any Thematic Link to an MonitoringFacility . The association has additional properties as defined in the association class AnyDomainLink .
Req 9	/req/req-class-MonitoringFacility/involvedIn MonitoringActivity(s) in which the MonitoringFacility is involved.

10.3 MonitoringNetwork

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringNetwork/MonitoringNetwork-sem Administrative or organisational grouping of EnvironmentalMonitoringFacilities managed the same way for a specific

	purpose, targeting a specific area. Each network respects common rules aiming at ensuring coherence of the observations, especially for purposes of EnvironmentalMonitoringFacilities , mandatory parameters selection, measurement methods and measurement regime.
Req 2	/req/req-class-MonitoringNetwork/contains A link pointing to the MonitoringFacility(s) included in this MonitoringNetwork . The association has additional properties as defined in the association class NetworkFacility .
Req 3	/req/req-class-MonitoringNetwork/observingCapability A link pointing to the explicit capability of an MonitoringNetwork . This provides a clean link between the observed property, the procedure used as well as the location of the measurement.
Req 4	/req/req-class-MonitoringNetwork/broader A link pointing to a broader MonitoringNetwork (a higher level in a hierarchical structure). The association has additional properties as defined in the association class Hierarchy .
Req 5	/req/req-class-MonitoringNetwork/narrower A link pointing to narrower MonitoringNetwork(s) (a lower level in a hierarchical structure). The association has additional properties as defined in the association class Hierarchy .
Req 6	/req/req-class-MonitoringNetwork/involvedIn MonitoringActivity(s) in which the AbstractMonitoringFeature is involved.

10.4 MonitoringActivity

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivity	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringActivity/MonitoringActivity-sem Specific set of MonitoringFacility(s) and/or MonitoringNetwork(s) used for a given domain in a coherent and concise timeframe, area and purpose. Usually, the information collected is treated as one time step in a long term monitoring programme. It is a concrete realisation of a given MonitoringProgramme .
Req 2	/req/req-class-MonitoringActivity/boundingBox

	Bounding box in which the MonitoringActivity takes place.
Req 3	/req/req-class-MonitoringActivity/usesFacility Specific set of EnvironmentalMonitoringFacilities involved in an MonitoringActivity .
Req 4	/req/req-class-MonitoringActivity/usesNetwork Specific set of MonitoringNetworks involved in an MonitoringActivity .
Req 5	/req/req-class-MonitoringActivity/setUpFor MonitoringProgramme(s) for which the MonitoringActivity is set up.

10.5 MonitoringProgramme

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramme	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringProgramme/MonitoringProgramme-sem Framework based on policy relevant documents defining the target of a collection of observations and/or the deployment of MonitoringFacility(s) and/or MonitoringNetwork(s) on the field. Usually, an MonitoringProgramme has a long term perspective over at least a few years.
Req 2	/req/req-class-MonitoringProgramme/observingCapability A link pointing to the explicit capability of an MonitoringProgramme . This provides a clean link between the observed property, the procedure used as well as the location of the measurement.
Req 3	/req/req-class-MonitoringProgramme/broader A link pointing to a broader MonitoringProgramme (a higher level in a hierarchical structure). The association has additional properties as defined in the association class Hierarchy .
Req 4	/req/req-class-MonitoringProgramme/narrower A link pointing to narrower MonitoringProgramme(s) (a lower level in a hierarchical structure). The association has additional properties as defined in the association class Hierarchy .
Req 5	/req/req-class-MonitoringProgramme/triggers A Link pointing to EnvironmentalMonitoringActivity(s) triggered by the EnvironmentalMonitoringProgramme .

11. Logical Monitoring Extended

11.1 General

In this section an extended version of the Monitoring model is presented, providing additional attributes stemming from the original INSPIRE Environmental Monitoring Facilities specification. This has been done to enable reuse of these extended attributes as required within specific scenarios; fine grained requirements classes enable mix and match of these extensions, as required to tailor custom solutions.

The core classes of this model, as shown in Figure 8, are:

- **MonitoringFacility:** A georeferenced object directly collecting or processing data about objects whose properties (e.g. physical, chemical, biological or other aspects of environmental conditions) are repeatedly observed or measured.
- **MonitoringNetwork:** Administrative or organisational grouping of EnvironmentalMonitoringFacilities managed the same way for a specific purpose, targeting a specific area.
- **MonitoringActivity:** Specific set of MonitoringFacilities or MonitoringNetworks used for a given domain in a coherent and concise timeframe, area and purpose. Usually, the information collected is treated as one time step in a long.term monitoring programme. It is a concrete realisation of a given MonitoringProgramme.
- **MonitoringProgramme:** Framework based on policy relevant documents defining the target of a collection of observations and/or the deployment of MonitoringFacilities or MonitoringNetworks on the field. Usually, a Monitoring Programme has a long-term perspective over at least a few years.

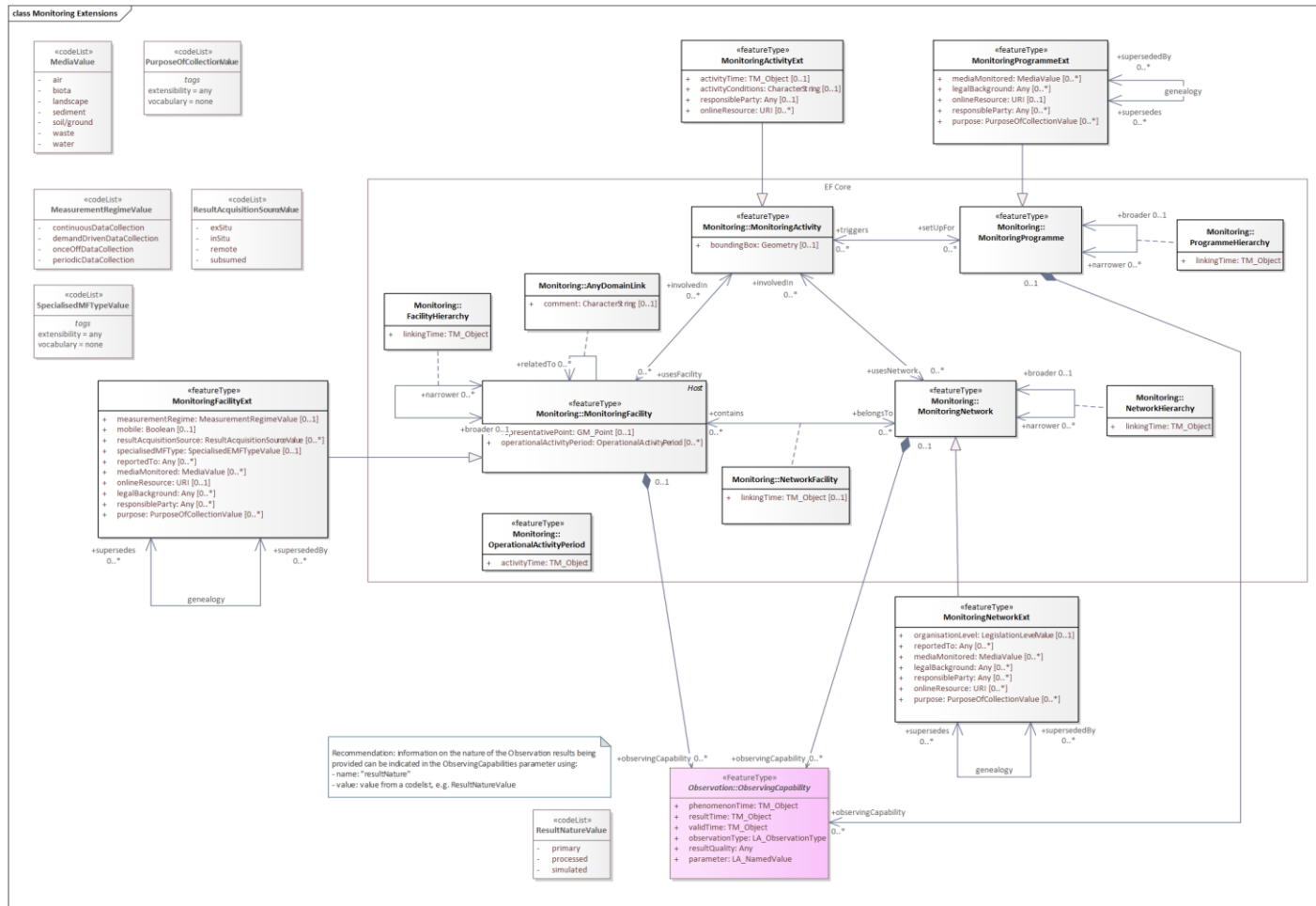


Figure 8: Extended Monitoring Schema

When utilizing the Monitoring Extensions in the creation of domain models, constraints can be useful to tailor the complexity of these types to specific use cases. In Figure 9, MyMonitoringFacility has been created based on MonitoringFacilityExt, adding a constraint mandating the provision of the mediaMonitored attribute.

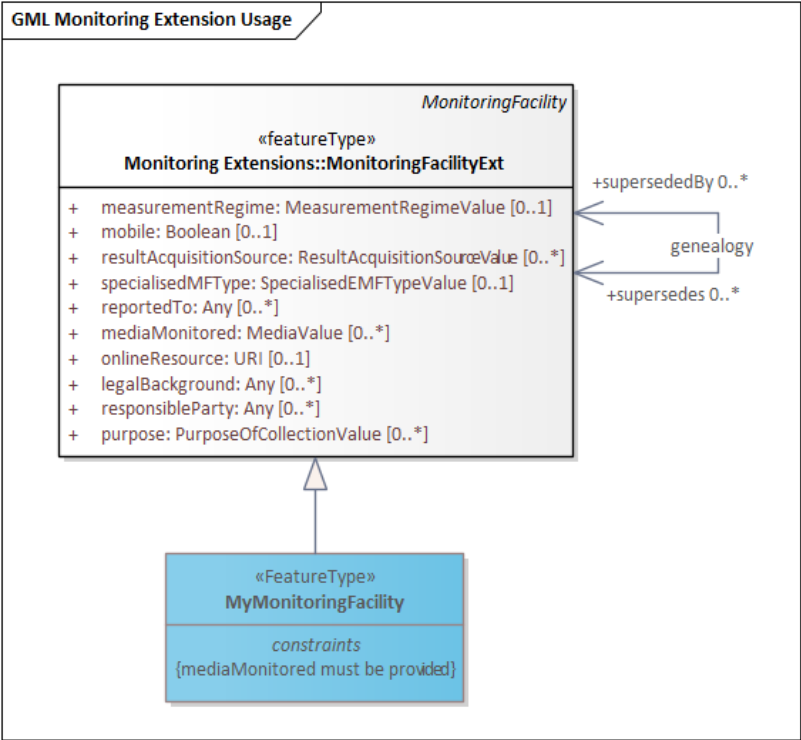


Figure 9: Using Extended Monitoring

11.2 MonitoringFacility Extended

11.2.1 reportedTo

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-reportedTo	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/reportedTo Information on the involvement of the MonitoringFacility in reporting.

11.2.2 mediaMonitored

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-mediaMonitored	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/mediaMonitored Monitored environmental medium.

11.2.3 legalBackground

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-legalBackground	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/legalBackground The legal context, in which the management and regulation of the MonitoringFacility is defined.

11.2.4 responsibleParty

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-responsibleParty	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/responsibleParty Responsible party for the MonitoringFacility .

11.2.5 onlineResource

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-onlineResource	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class

Req 1	/req/req-class-MonitoringFacility/onlineResource A link to an external document providing further information on the MonitoringFacility .
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11.2.6 purpose

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-purpose	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/purpose Reason for which the MonitoringFacility has been generated.

11.2.7 measurementRegime

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-measurementRegime	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/measurementRegime Regime of the measurement.

11.2.8 mobile

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-mobile	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/mobile Indicate whether the MonitoringFacility is mobile (repositionable) during the acquisition of the observation.

11.2.9 resultAcquisitionSource

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-resultAcquisitionSource	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/resultAcquisitionSource Source of result acquisition

11.2.10 specialisedMFTType

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-specialisedMFTType	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Dependency	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SpecialisedMFTValue
Req 1	/req/req-class-MonitoringFacility/specialisedMFTType Categorisation of MonitoringFacility (s) generally used by domain and in national settings.

11.2.11 geometryRequired

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-geometryRequired	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/geometryRequired Geometry and representativePoint cannot both be empty.

11.2.12 supersedes

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-supersedes	
Target type	Logical Model

Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/supersedes In a genealogy, the MonitoringFacility(s) that has(have) been deactivated/replaced by another one.

11.2.13 supersededBy

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-supersededBy	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringFacility/supersededBy In a genealogy, the newly active MonitoringFacility(s) that replaces(replace) the superseded one.

11.3 MonitoringNetwork Extended

11.3.1 mediaMonitored

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork-mediaMonitored	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringNetwork/mediaMonitored Monitored environmental medium.

11.3.2 legalBackground

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork-legalBackground	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringNetwork/legalBackground

	The legal context, in which the management and regulation of the MonitoringNetwork is defined.
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11.3.3 responsibleParty

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork-responsibleParty	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringNetwork/responsibleParty Responsible party for the MonitoringNetwork.

11.3.4 onlineResource

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork-onlineResource	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringNetwork/onlineResource A link to an external document providing further information on the MonitoringNetwork

11.3.5 purpose

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork-purpose	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringNetwork/purpose Reason for which the MonitoringNetwork has been generated.

11.3.6 organisationLevel

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork-organisationLevel	

Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringNetwork/organisationLevel Level of legal organisation the MonitoringNetwork is affiliated with.

11.3.7 supersedes

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork-supersedes	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringNetwork/supersedes In a genealogy, the MonitoringNetwork(s) that has(have) been deactivated/replaced by another one.

11.3.8 supersededBy

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork-supersededBy	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringNetwork/supersededBy In a genealogy, the newly active MonitoringNetwork(s) that replaces(replace) the superseded one.

11.4 MonitoringActivity Extended

11.4.1 activityTime

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivity-activityTime	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class

Req 1	/req/req-class-MonitoringActivity/activityTime Lifespan of the MonitoringActivity .
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11.4.2 activityConditions

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivity-activityConditions	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringActivity/activityConditions Textual description of the MonitoringActivity .

11.4.3 responsibleParty

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivity-responsibleParty	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringActivity/responsibleParty Responsible party for the MonitoringActivity .

11.4.4 onlineResource

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivity-onlineResource	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringActivity/onlineResource A link to an external document providing further information on the MonitoringActivity .

11.5 MonitoringProgramme Extended

11.5.1 mediaMonitored

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramme-mediaMonitored	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringProgramme/mediaMonitored Monitored environmental medium.

11.5.2 legalBackground

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramme-legalBackground	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringProgramme/legalBackground The legal context, in which the management and regulation of the MonitoringProgramme is defined.

11.5.3 responsibleParty

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramme-responsibleParty	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringProgramme/responsibleParty Responsible party for the MonitoringProgramme .

11.5.4 onlineResource

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramme-onlineResource	
Target type	Logical Model

Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringProgramme/onlineResource A link to an external document providing further information on the MonitoringProgramme .

11.5.5 purpose

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramme-purpose	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringProgramme/purpose Reason for which the MonitoringProgramme has been generated.

11.5.6 supersedes

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramme-supersedes	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringProgramme/supersedes In a genealogy, the MonitoringProgramme(s) that has(have) been deactivated/replaced by another one.

11.5.7 supersededBy

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramme-supersededBy	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MonitoringProgramme/supersededBy In a genealogy, the newly active MonitoringProgramme(s) that replaces(replace) the superseded one.

11.6 Codelists

11.6.1 Codelist SpecialisedMFTypeValue

An empty extension-point for providing various classification schemes for **MonitoringFacilities**.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SpecialisedMFTypeValue	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-SpecialisedMFTypeValue/SpecialisedMFTypeValue-sem An empty extension-point for providing various classification schemes for MonitoringFacilities , generally used by domain and in national settings. EXAMPLES: platform, site, station, sensor,

11.6.2 Codelist PurposeOfCollectionValue

An empty extension-point for providing various classification schemes describing the reason for which **MonitoringFacilities**, **MonitoringNetworks** or **MonitoringProgrammes** have been created.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PurposeOfCollectionValue	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-PurposeOfCollectionValue/PurposeOfCollectionValue-sem An empty extension-point for providing various classification schemes describing the reason for which MonitoringFacilities , MonitoringNetworks or MonitoringProgrammes have been created. EXAMPLES: European reporting obligation, international collaboration, long term monitoring of contaminated sites,

11.6.3 Codelist MediaValue

A codelist for classifying **MonitoringFacilities**, **MonitoringNetworks** or **MonitoringProgrammes** by the media they monitor.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MediaValue	
Target type	Logical Model

Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MediaValue/MediaValue-sem A codelist for classifying MonitoringFacilities, MonitoringNetworks or MonitoringProgrammes by the media they monitor. The following entries shall be provided: • air • biota • landscape • sediment • soil/ground • waste • water

11.6.4 Codelist MeasurementRegimeValue

A codelist for classifying **MonitoringFacilities** by the measurement regime guiding their operation.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MeasurementRegimeValue	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-MeasurementRegimeValue/MeasurementRegimeValue-sem A codelist for classifying MonitoringFacilities by the measurement regime guiding their operation. The following entries shall be provided: • continuousDataCollection: Data is collected on a continuous basis. There is usually no end date, as further data is collected. • demandDrivenDataCollection: Data is collected on demand. • onceOffDataCollection: Data is collected only once in this configuration. No further observations in this configuration can be expected. • periodicDataCollection: Data is collected at regular intervals. No information is available at to the data collection interval.

11.6.5 Codelist ResultAcquisitionSourceValue

A codelist for classifying **MonitoringFacilities** by their result acquisition source.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ResultAcquisitionSourceValue	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-ResultAcquisitionSourceValue/ResultAcquisitionSourceValue-sem A codelist for classifying MonitoringFacilities by their result acquisition source. The following entries shall be provided: • exSitu: The FeatureOfInterest is a specimen taken from the ultimate FeatureOfInterest (i.e. the sampledFeature). • inSitu: The FeatureOfInterest is a sampling feature which is co-located with the ultimate FeatureOfInterest (i.e. the sampledFeature). • remote: The FeatureOfInterest is a sampling feature which is also the ultimate FeatureOfInterest (i.e. the sampledFeature). • subsumed: The value is inherited from children.

11.6.6 Codelist ResultNatureValue

A codelist for classifying **MonitoringFacilities**, **MonitoringNetworks** or **MonitoringProgrammes** by the nature of their results.

Requirements Class	
http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ResultNatureValue	
Target type	Logical Model
Dependency	ISO 19103:2015 Geographic information – Conceptual schema language, UML2 conformance class
Req 1	/req/req-class-ResultNatureValue/ResultNatureValue-sem A codelist for classifying MonitoringFacilities, MonitoringNetworks or MonitoringProgrammes by the nature of their results. The following entries shall be provided:

	• primary: The result provided with the observation is the direct result of an estimate of a property on the featureOfInterest. No further processing has been performed. Processing may have taken place, but only in the sense of the measurement methodology itself, i.e. converting the millivolt returned from the sensor to the concentration of a substance. • processed: The result provided, while usually based on primary measurements, has been substantially processed. This processing can be of diverse natures, in some situations complex aggregates are provided, in other situations, the existing values are interpolated to a continuum. • simulated: The result provided, while usually based on primary measurements, is based on an interpretation model, and provides a simulation of past or future states of the media being analyzed. In this case, the existing values are usually extrapolated into the past or future.
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Annex

Annex A: Conformance Class Test Suite (Normative)

Conformance class: Observation

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Observation		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Observation	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Observation/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservableProperty

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservableProperty		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservableProperty	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservableProperty/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservingProcedure

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservingProcedure	
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservingProcedure

Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservingProcedure/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: Observer

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Observer		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Observer	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Observer/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: Host

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Host		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Host	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Host/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: Deployment

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Deployment		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Deployment	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Deployment/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservationCharacteristics

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCharacteristics		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCharacteristics	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCharacteristics/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservationCollection

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCollection		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCollection	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCollection/test-1	

	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: NamedValue

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-NamedValue		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-NamedValue	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-NamedValue/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservationType

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationType		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationType	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationType/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservationTypeByResultType

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationTypeByResultType		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationTypeByResultType	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationTypeByResultType/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: ObservationCollectionType

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCollectionType		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-ObservationCollectionType	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-ObservationCollectionType/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Sampling

Conformance class: Sample

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sample	
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sample

Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sample/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SpatialSample

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SpatialSample		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SpatialSample	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SpatialSample/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MaterialSample

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MaterialSample		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MaterialSample	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MaterialSample/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: StatisticalSample

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-StatisticalSample		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-StatisticalSample	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-StatisticalSample/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: Sampling

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sampling		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sampling	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sampling/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: Sampler

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sampler		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-Sampler	
	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-Sampler/test-1	

Test	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SamplingProcedure

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SamplingProcedure		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SamplingProcedure	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SamplingProcedure/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: PreparationStep

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PreparationStep		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PreparationStep	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PreparationStep/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: PreparationProcedure

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PreparationProcedure		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PreparationProcedure	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PreparationProcedure/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SampleCollection

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleCollection		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleCollection	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleCollection/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: PhysicalDimension

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PhysicalDimension	
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-PhysicalDimension

Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-PhysicalDimension/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: NamedLocation

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-NamedLocation		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-NamedLocation	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-NamedLocation/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: StatisticalClassification

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-StatisticalClassification		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-StatisticalClassification	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-StatisticalClassification/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SampleType

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleType		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleType	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleType/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SampleTypeByGeometryType

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleTypeByGeometryType		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SampleTypeByGeometryType	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SampleTypeByGeometryType/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: SamplerType

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SamplerType	
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-SamplerType

Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-SamplerType/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

EF

Conformance class: MonitoringFacility

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacility		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacility	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacility/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-geometryRequired

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacility-geometryRequired		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-geometryRequired	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-geometryRequired/test-1	

	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetwork

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetwork		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetwork	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetwork/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringActivity

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivity		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivity	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivity/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgramme

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramme		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgramme	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgramme/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

EF Extended

MonitoringFacilityExt

Conformance class: MonitoringFacilityExt-reportedTo

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-reportedTo		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-reportedTo	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-reportedTo/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-additionalDescription

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-additionalDescription		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-additionalDescription	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-additionalDescription/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-mediaMonitored

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-mediaMonitored		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-mediaMonitored	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-mediaMonitored/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-legalBackground

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-legalBackground	
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-legalBackground

Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-legalBackground/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: **MonitoringFacilityExt-responsibleParty**

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-responsibleParty		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-responsibleParty	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-responsibleParty/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: **MonitoringFacilityExt-onlineResource**

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-onlineResource		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-onlineResource	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-onlineResource/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

	Test method	Inspect the documentation of the application, schema or profile.
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Conformance class: MonitoringFacilityExt-purpose

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-purpose		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-purpose	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-purpose/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-measurementRegime

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-measurementRegime		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-measurementRegime	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-measurementRegime/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-mobile

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-mobile		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-mobile	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-mobile/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-resultAcquisitionSource

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-resultAcquisitionSource		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-resultAcquisitionSource	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-resultAcquisitionSource/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-specialisedMFType

Conformance Class	
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-specialisedMFType	
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-specialisedMFType

Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-specialisedMFTType/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-supersedes

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-supersedes		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-supersedes	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-supersedes/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringFacilityExt-supersededBy

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-supersededBy		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringFacilityExt-supersededBy	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringFacilityExt-supersededBy/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

	Test method	Inspect the documentation of the application, schema or profile.
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MonitoringNetwork

Conformance class: MonitoringNetworkExt-additionalDescription

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-additionalDescription		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-additionalDescription	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-additionalDescription/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-mediaMonitored

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-mediaMonitored		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-mediaMonitored	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-mediaMonitored/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-legalBackground

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-legalBackground		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-legalBackground	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-legalBackground/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-responsibleParty

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-responsibleParty		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-responsibleParty	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-responsibleParty/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-onlineResource

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-onlineResource		

Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-onlineResource	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-onlineResource/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-purpose

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-purpose		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-purpose	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-purpose/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringNetworkExt-organisationLevel

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-organisationLevel		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-organisationLevel	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-organisationLevel/test-1	

	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: **MonitoringNetworkExt-supersedes**

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-supersedes		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-supersedes	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-supersedes/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: **MonitoringNetworkExt-supersededBy**

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-supersededBy		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringNetworkExt-supersededBy	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringNetworkExt-supersededBy/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

MonitoringActivity

Conformance class: MonitoringActivityExt-activityTime

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-activityTime		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-activityTime	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-activityTime/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringActivityExt-activityConditions

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-activityConditions		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-activityConditions	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-activityConditions/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringActivityExt-responsibleParty

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-responsibleParty		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-responsibleParty	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-responsibleParty/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringActivityExt-onlineResource

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-onlineResource		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringActivityExt-onlineResource	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringActivityExt-onlineResource/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

MonitoringProgramme

Conformance class: MonitoringProgrammeExt-additionalDescription

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-additionalDescription		

Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgrammeExt-additionalDescription	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-additionalDescription/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgrammeExt-mediaMonitored

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-mediaMonitored		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgrammeExt-mediaMonitored	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-mediaMonitored/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgrammeExt-legalBackground

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-legalBackground		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgrammeExt-legalBackground	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-legalBackground/test-1	

	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgrammeExt-responsibleParty

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-responsibleParty		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgrammeExt-responsibleParty	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-responsibleParty/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgrammeExt-onlineResource

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-onlineResource		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgrammeExt-onlineResource	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-onlineResource/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgrammeExt-purpose

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-purpose		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgrammeExt-purpose	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-purpose/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgrammeExt-supersedes

Conformance Class		
http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-supersedes		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgrammeExt-supersedes	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-supersedes/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Conformance class: MonitoringProgrammeExt-supersededBy

Conformance Class

http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-supersededBy		
Requirement	http://www.opengis.net/spec/OMS-L/1.0/req/req-class-MonitoringProgrammeExt-supersededBy	
Test	http://www.opengis.net/spec/OMS-L/1.0/conf/conf-class-MonitoringProgrammeExt-supersededBy/test-1	
	Test purpose	Verify that all requirements from the requirements class have been fulfilled.
	Test method	Inspect the documentation of the application, schema or profile.

Annex <insert annex number>: Bibliography

<A Bibliography, if present, shall appear as the last annex. >